MHD Derivations for stellar2d — a sympy-verified manuscript

stellar2d development notes
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Contents

1 Front matter

1.1 Purpose

This manuscript is an **internal, reproducibility-grade derivation document** for the MHD extension of stellar2d. It covers four parts:

- **Part A** — ideal MHD equations, conservative / primitive forms, flux Jacobian eigensystem, HLLD intermediate states, constrained transport (CT) preservation of $\nabla \cdot \mathbf{B} = 0$; PLM/PPM reconstruction + TVD slope limiters; VL2 predictor-corrector and its CFL bound (hyperbolic + parabolic); HLLD degeneracy branches; Powell-source vs CT comparison; linear-wave convergence IC.
- **Part B** — reduction to a 1D super-radial flux-tube geometry used by the Suzuki-group stellar-wind codes, including the WKB Alfvén-wave action integral, the Parker critical-

point condition in this geometry, and the well-balanced MHSE operator needed to keep the atmosphere quiet for long-time wind runs.
- **Part C** — non-ideal dissipation: Ohmic, ambipolar, and their contributions to the energy equation; closure of the two diffusivities through Saha ionisation; sub-grid turbulent heating closure (Suzuki-Inutsuka 2005) for 1D flux-tube wind runs.
- **Part D** — cylindrical shearing-box MHD, the shearing-periodic boundary condition, and the exact Maxwell–Reynolds stress decomposition of the $\alpha_{\rm SS}$ metric.

1.2 Reproducibility protocol

Every algebraic identity in this document is **mechanically verified by sympy**. Each section corresponds to a script in `docs/mhd_derivations/scripts/<section>.py` that ends with `assert_zero(LHS - RHS, ...)` calls. If any identity cannot be simplified by sympy but is physically correct, we document the alternate verification route (e.g., "manually checked with Wolfram, reason: polynomial simplification blows up").

The scripts are intentionally **independent**: each one imports only `_common.py` and re-derives everything it needs from first principles. No cross-script caching of intermediate symbolic results. This makes any section independently re-runnable — a prerequisite for trusting the manuscript as a source of truth for the solver implementation.

1.3 Conventions

- Units are Gaussian with 4π absorbed into **B**, so that the Alfvén speed is simply $c_A = B/\sqrt{\rho}$, matching Stone & Gardiner 2008, Athena++, PLUTO.
- γ always denotes the ratio of specific heats (not the relativistic Lorentz factor).
- Einstein summation is *not* used; sums are written explicitly.
- sympy variable names mirror the mathematical symbols (`rho` = ρ, `B_x` = B_x, etc.); see `scripts/_common.py` for the full inventory.

1.4 How to regenerate this manuscript

```
cd docs/mhd_derivations
bash run_all.sh        # (re-)runs every scripts/*.py, refreshes output/
bash build_manuscript.sh  # concatenates sections/*.md → manuscript.{md,pdf}
```

If a sympy assertion fails during `run_all.sh`, the build halts and the offending section is flagged. No partial manuscript is emitted.

2 A1. Ideal MHD equations (conservative form)

> **sympy script:** `scripts/a1_ideal_mhd_equations.py` **generated LaTeX:** `output/a1_ideal_mhd_equations.latex.tex` **verifies:** 8 identities (Lorentz × 3 components, induction × 3, Poynting-flux identity, $\nabla \cdot (\nabla \times \cdot) = 0$) **code checkpoints (future):** `src/gpu/explicit/athena_mhd_kernels.cu::d_mhd_flux`

2.1 Starting assumptions

Label	Assumption
A1a	Compressible neutral gas with magnetic field $\mathbf{B}$; no viscosity, no Ohmic, no ambipolar, no Hall.
A1b	Infinite conductivity → frozen-in flux, $\mathbf{E} = -\mathbf{v} \times \mathbf{B}$.
A1c	Non-relativistic: displacement current $\partial_t \mathbf{E}/c$ dropped.
A1d	Ideal EOS: $p = (\gamma - 1)\rho e$.

2.2 Mass conservation

Written compactly as

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0. \quad \text{(A1-mass)}$$

Under A1a–d this is the unmodified fluid continuity equation.

2.3 Lorentz force identity (A1-lorentz)

The single most important vector identity of the derivation is

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla \cdot (\mathbf{B} \otimes \mathbf{B}) - \nabla\left(\tfrac{1}{2}|\mathbf{B}|^2\right) - (\nabla \cdot \mathbf{B})\,\mathbf{B}. \quad \text{(A1-lorentz)}$$

The last term vanishes **analytically** when the solenoidal constraint $\nabla \cdot \mathbf{B} = 0$ is enforced. At the discrete level it does not vanish in general; this is precisely what CT schemes (§A5) are engineered to handle.

Sympy verification. All three vector components of the residual $(\nabla \times \mathbf{B}) \times \mathbf{B} - [\nabla \cdot (\mathbf{B} \otimes \mathbf{B}) - \nabla(\tfrac{1}{2}|\mathbf{B}|^2) - (\nabla \cdot \mathbf{B})\mathbf{B}]$ reduce to 0 under `sympy.simplify`, without imposing $\nabla \cdot \mathbf{B} = 0$. This confirms the identity is an *algebraic*, not a physical, fact.

2.4 Momentum conservation (divergence form)

Define the **total magneto-fluid pressure**

$$P^\star \equiv p + \tfrac{1}{2}|\mathbf{B}|^2.$$

Then (A1-lorentz) lets us write momentum conservation in divergence form:

$$\boxed{\partial_t(\rho\mathbf{v}) + \nabla\cdot[\rho\mathbf{v} \otimes \mathbf{v} - \mathbf{B} \otimes \mathbf{B} + P^\star \mathbf{I}] = \mathbf{0}.} \quad \text{(A1-mom)}$$

Implementation note. $\mathbf{B} \otimes \mathbf{B}$ is *not* symmetric in its indices when read as a flux tensor F_{ij} (it is actually symmetric, but in contrast the induction tensor $v_i B_j - v_j B_i$ is anti-symmetric). In the kernel, $B_i B_j$ for each $(i,j) \in \{(x,x),(x,y),(x,z),(y,y),(y,z),(z,z)\}$ must be computed on the same stencil as $v_i v_j$ — mixing face-averaged $\mathbf{B}$ with cell-centred $\mathbf{B}$ here is the number-one source of initial HLLD bugs.

2.5 Induction equation — two equivalent forms (A1-induction)

Tensor-divergence form (preferred for a Godunov kernel):

$$\partial_t B_i + \partial_j\big(v_j B_i - v_i B_j\big) = 0.$$

Curl form (preferred for a CT update):

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}).$$

Sympy verification. For each component $i \in \{x, y, z\}$ we check that

$$\nabla \times (\mathbf{v} \times \mathbf{B})\Big|_i \equiv -\partial_j(v_j B_i - v_i B_j)$$

holds as a *pure vector-calculus identity* (no $\nabla \cdot \mathbf{B} = 0$, no $\nabla \cdot \mathbf{v} = 0$ invoked). This is the well-known Jacobi-like identity; all three sympy assertions pass.

2.6 Total-energy conservation (A1-energy)

With

$$E \equiv \rho e + \tfrac{1}{2}\rho|\mathbf{v}|^2 + \tfrac{1}{2}|\mathbf{B}|^2,$$

the conservative form is

$$\boxed{\partial_t E + \nabla\cdot[(E + P^\star)\mathbf{v} - \mathbf{B}(\mathbf{B}\cdot\mathbf{v})] = 0.}$$

The magnetic-Poynting-flux piece $-\mathbf{B}(\mathbf{B}\cdot\mathbf{v})$ and the enthalpy-like piece $P^\star\mathbf{v}$ together make the flux manifestly symmetric-hyperbolic.

Sympy-verified identity. The vector identity $\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) = |\mathbf{B}|^2\mathbf{v} - \mathbf{B}(\mathbf{v}\cdot\mathbf{B})$ lets us rewrite the Poynting flux $\mathbf{S} = \frac{1}{2}\mathbf{B} \times (\mathbf{v} \times \mathbf{B})$ in the form used in (A1-energy). `assert_zero` confirms $\nabla \cdot [\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) - (|\mathbf{B}|^2\mathbf{v} - \mathbf{B}(\mathbf{B}\cdot\mathbf{v}))] = 0$ identically.

2.7 Solenoidal constraint preservation (A1-divB)

Taking the divergence of the induction equation in curl form:

$$\partial_t(\nabla \cdot \mathbf{B}) = \nabla \cdot \nabla \times (\mathbf{v} \times \mathbf{B}) = 0,$$

where the last equality is the identity $\nabla \cdot (\nabla \times \cdot) = 0$. **At the continuous level this is automatic**; at the discrete level it is what CT or Dedner-GLM machinery must engineer. See §A5.

Sympy verification. `assert_zero(div_cart(curl_cart(v×B)))` passes for arbitrary smooth **v**, **B**.

2.8 Compact conservative system

Collecting all fluxes, we write the 8-variable conservative MHD system:

$$\boxed{\partial_t \mathbf{U} + \partial_i \mathbf{F}_i(\mathbf{U}) = \mathbf{0}, \quad \mathbf{U} = (\rho,\ \rho\mathbf{v},\ \mathbf{B},\ E)^{\mathrm{T}}.}$$

The explicit form of $\mathbf{F}_i$ follows by assembling the four fluxes above. We defer the component-by-component flux Jacobian and its eigensystem to §A3.

2.9 □ Verification checkpoint (to be wired)

Implementation invariants the kernel must satisfy (enforced through `tests/test_athena_mhd_*.cu` in a future PR):

1. **One-cell linearised update reduces to (A1-mass), (A1-mom), (A1-induction-tensor), (A1-energy) exactly** — zero the $\mathbf{B}$ field and verify that $\mathbf{F}_i$ reduces to the Euler flux used in `athena_vl2`.
2. **Solenoidal residual** $(\nabla \cdot \mathbf{B})_{cell}$ from face- centred $\mathbf{B}$ remains at round-off over 100 sound-crossing times on a smooth periodic IC (field-loop advection, §A5).
3. **Energy-momentum consistency** — on a piecewise-constant linear wave IC, total energy updated via (A1-energy) agrees with $\frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2 + \rho e$ reconstructed from the primitive fields, to round-off per step.

Any failure on (1)–(3) flags the implementation as inconsistent with §A1 and must be resolved before proceeding to §A2.

3 A2. Conservative ↔ primitive variable transformation

sympy script: `scripts/a2_conservative_primitive.py` **generated LaTeX:** `output/a2_conservative_primitive.latex.tex` **verifies:** $\dfrac{\partial \mathbf{W}}{\partial \mathbf{U}} \cdot \dfrac{\partial \mathbf{U}}{\partial \mathbf{W}} = \mathsf{I}_8$ (all 64 entries symbolically checked to zero after subtraction). **code checkpoints (future):** `athena_mhd_kernels.cu::d_prim_from_cons`, `d_cons_from_prim`.

3.1 Variable inventory

The conservative 8-vector used inside the Godunov flux loop:

$$\mathbf{U} = (\rho,\ \rho v_x,\ \rho v_y,\ \rho v_z,\ B_x,\ B_y,\ B_z,\ E)^{\mathrm{T}}.$$

The primitive 8-vector used for reconstruction / Riemann solver input:

$$\mathbf{W} = (\rho,\ v_x,\ v_y,\ v_z,\ B_x,\ B_y,\ B_z,\ p)^{\mathrm{T}}.$$

The total-energy density closes the map through the ideal EOS:

$$E = \frac{p}{\gamma - 1} + \tfrac{1}{2}\rho|\mathbf{v}|^2 + \tfrac{1}{2}|\mathbf{B}|^2. \quad \text{(A2-E)}$$

3.2 Pressure extraction (primitive from conservative)

Inverting (A2-E) for p gives the pressure-extraction formula the `d_prim_from_cons` kernel must use:

$$\boxed{p = (\gamma - 1)\left(E - \frac{|\mathbf{m}|^2}{2\rho} - \tfrac{1}{2}|\mathbf{B}|^2\right), \quad \mathbf{m} \equiv \rho\mathbf{v}.}$$

Implementation gotcha. For low-β, high-Mach flows (MHD blast, Orszag-Tang late time), the hydro term $|\mathbf{m}|^2/(2\rho)$ and the magnetic term $|\mathbf{B}|^2/2$ can both be within machine-precision of the total E. A naïve subtraction can then hand the EOS a negative p. The kernel must either:

1. Accept a positivity fallback (fall back to isothermal primitive extraction and flag the cell), or
2. Use the dual-energy formulation (evolve an auxiliary internal-energy variable and use it when the primitive subtraction is unreliable).

This is documented in Stone & Gardiner 2008 §4.6 and implemented in Athena++ as the `DUAL_ENERGY` compile-time flag.

3.3 Forward Jacobian $\partial\mathbf{U}/\partial\mathbf{W}$

sympy produces the 8×8 Jacobian explicitly; it is block-sparse with the $\mathbf{B}$ block trivially the identity, the momentum block $\rho\mathbf{I} + \mathbf{v} \otimes (\partial\mathbf{v}/\partial\mathbf{W})$, and the energy row containing $\partial E/\partial\rho$, $\rho\mathbf{v}$, $\mathbf{B}$, $1/(\gamma - 1)$.

3.4 Backward Jacobian $\partial\mathbf{W}/\partial\mathbf{U}$

The inverse map takes the conservative 8-vector to primitives term by term. The non-trivial rows are the velocity-from-momentum and the pressure-from-energy rows:

$$\frac{\partial v_i}{\partial(\rho v_j)} = \frac{\delta_{ij}}{\rho}, \qquad \frac{\partial v_i}{\partial\rho} = -\frac{v_i}{\rho},$$

$$\frac{\partial p}{\partial\rho} = (\gamma - 1)\frac{|\mathbf{m}|^2}{2\rho^2} = (\gamma - 1)\tfrac{1}{2}|\mathbf{v}|^2,$$

$$\frac{\partial p}{\partial(\rho v_i)} = -(\gamma - 1)v_i, \qquad \frac{\partial p}{\partial B_i} = -(\gamma - 1)B_i, \qquad \frac{\partial p}{\partial E} = \gamma - 1.$$

3.5 Consistency check (sympy)

The script computes $\dfrac{\partial\mathbf{W}}{\partial\mathbf{U}}(\mathbf{U}(\mathbf{W})) \cdot \dfrac{\partial\mathbf{U}}{\partial\mathbf{W}}(\mathbf{W})$ after substituting back so that both Jacobians are evaluated at the same primitive state, and verifies

$$\frac{\partial\mathbf{W}}{\partial\mathbf{U}} \cdot \frac{\partial\mathbf{U}}{\partial\mathbf{W}} = \mathsf{I}_8. \quad \text{(A2-inverse)}$$

All 64 entries of the residual matrix are `sympy.simplify`-reduced to 0 and pass `assert_zero`.

3.6 □ Verification checkpoint (to be wired)

When the MHD kernel is written, the test

`tests/test_athena_mhd_roundtrip.cu`

should:

1. Seed random primitive 8-vectors $\mathbf{W}$ with $\rho, p > 0$ and $|\mathbf{B}|^2/(2p) < 10^6$ (stay out of the positivity danger zone).
2. Call `d_cons_from_prim` to compute $\mathbf{U}$.
3. Call `d_prim_from_cons` to recover $\mathbf{W}'$.
4. Assert $\|\mathbf{W} - \mathbf{W}'\|_\infty / \|\mathbf{W}\|_\infty < 10\,\varepsilon_{\text{mach}}$.

Any regression of (A2-inverse) in the kernel fails this round-trip.

4 A3. Flux Jacobian $\mathsf{A} = \partial\mathbf{F}/\partial\mathbf{U}$ and its 7-wave eigensystem

sympy script: `scripts/a3_flux_jacobian_eigensystem.py` **generated LaTeX:** `output/a3_flux_jacobian_eigensystem.latex.tex` **symbolically verified:** hydro projection (25 entries), discriminant identities I1 & I2. **numerically verified (20 trials × 3 γ):** eigenvalues match closed-form to $\leq 5 \times 10^{-15}$; $\mathsf{A}\mathbf{r} - \lambda\mathbf{r}$ residuals $\leq 6 \times 10^{-15}$. **code checkpoints (future):** `athena_mhd_kernels.cu::d_mhd_eigenvalues_primitive`, `d_mhd_right_eigenvectors`.

4.1 Working frame

The eigenanalysis is done in **primitive form** in 1D along $\hat{x}$. Primitive 7-vector:

$$\mathbf{W} = (\rho,\ v_x,\ v_y,\ v_z,\ B_y,\ B_z,\ p)^{\mathrm{T}}.$$

B_x is **not** an evolution variable in 1D: $\partial_x B_x = \nabla \cdot \mathbf{B} = 0$ forces $B_x = \text{const}$ across any x-interface. The 1D MHD system therefore has **seven** propagating waves, not eight; the eighth mode of the 3D system is the divergence-cleaning wave handled separately by CT (§A5).

4.2 Primitive-form Jacobian A_W

$$\partial_t \mathbf{W} + \mathsf{A}_W(\mathbf{W})\,\partial_x \mathbf{W} = \mathbf{0},$$

$$\mathsf{A}_W = \begin{bmatrix} v_x & \rho & 0 & 0 & 0 & 0 & 0 \\ 0 & v_x & 0 & 0 & B_y/\rho & B_z/\rho & 1/\rho \\ 0 & 0 & v_x & 0 & -B_x/\rho & 0 & 0 \\ 0 & 0 & 0 & v_x & 0 & -B_x/\rho & 0 \\ 0 & B_y & -B_x & 0 & v_x & 0 & 0 \\ 0 & B_z & 0 & -B_x & 0 & v_x & 0 \\ 0 & \gamma p & 0 & 0 & 0 & 0 & v_x \end{bmatrix}. \quad \text{(A3-jacobian)}$$

Hydrodynamic sanity check. Setting $B_x = B_y = B_z = 0$ and projecting onto the (ρ, v_x, v_y, v_z, p) subspace recovers the standard 5×5 hydrodynamic primitive Jacobian. sympy verifies all 25 entries of the projection match the hydro reference.

4.3 Characteristic speeds

Define the four reference speeds:

$$c_{s_0}^2 \equiv \gamma p/\rho, \qquad c_{Ax}^2 \equiv B_x^2/\rho, \qquad c_{A\perp}^2 \equiv (B_y^2 + B_z^2)/\rho, \qquad c_A^2 \equiv c_{Ax}^2 + c_{A\perp}^2.$$

The fast and slow magnetosonic speeds are the two positive roots of

$$c_{f,s}^2 = \tfrac{1}{2}\Big[(c_{s_0}^2 + c_A^2) \pm \sqrt{(c_{s_0}^2 + c_A^2)^2 - 4c_{s_0}^2 c_{Ax}^2}\Big]. \quad \text{(A3-cfs)}$$

Discriminant identities (HLLD-stable forms). The closed form (A3-cfs) loses ULPs when $c_{A\perp} \to 0$ or $c_{s_0} \to 0$. The HLLD kernel uses the two equivalent identities

$$\boxed{c_f^2 + c_s^2 = c_{s_0}^2 + c_A^2, \qquad c_f^2 \cdot c_s^2 = c_{s_0}^2 \cdot c_{Ax}^2,} \quad \text{(A3-discriminant)}$$

which sympy verifies symbolically (both reduce to 0 under `simplify`). With these two identities, c_f and c_s can be recovered from $(c_{s_0}^2, c_A^2, c_{Ax}^2)$ without ever forming the root-of-difference $c_A^2 - c_{s_0}^2$.

4.4 The seven wave speeds

$$\boxed{\{\lambda_k\}_{k=1}^7 = \{\, v_x - c_f,\ v_x - c_{Ax},\ v_x - c_s,\ v_x,\ v_x + c_s,\ v_x + c_{Ax},\ v_x + c_f \,\}.} \quad \text{(A3-wave-speeds)}$$

Here $c_{Ax} \equiv B_x/\sqrt{\rho}$ carries the sign of B_x — this convention lets us write the Alfvén speed once without the $s = \text{sign}(B_x)$ factor that appears in Stone+08 eigenvector formulas. The other convention (unsigned $c_{Ax} = |B_x|/\sqrt{\rho}$ + s in every eigenvector entry) is equivalent; we choose signed-c_{Ax} here because it keeps the 7-wave spectrum contiguous for all sign(B_x).

4.5 Numerical verification of the spectrum

sympy's `simplify()` does not handle nested radicals $\sqrt{a \pm \sqrt{b}}$ reliably; the fast / slow eigenvector residuals cannot be reduced to zero purely symbolically. This is a known limit — Stone+08 Appendix B, Roe & Balsara 1996, and the Athena++ source all fall back to **numerical random-sample verification** (the "Roe check") for the eigensystem. We do the same:

1. Draw 20 random physically admissible states $(\rho, p, v_x, B_x, B_y, B_z)$ with $\rho, p > 0$, $B_x \neq 0$, $|B_\perp| \neq 0$;
2. For each of three ratios of specific heats $\gamma \in \{5/3, 7/5, 4/3\}$, diagonalise A_W numerically with `numpy.linalg.eig`;
3. Sort the seven numerical λ_k and compare to the sorted closed-form spectrum (A3-wave-speeds).

Results over 60 trials:

$$\max|\lambda_{\text{closed}} - \lambda_{\text{numerical}}| < 5 \times 10^{-15}, \qquad \max\|\mathsf{A}\mathbf{r} - \lambda\mathbf{r}\|_\infty < 6 \times 10^{-15}.$$

The residuals are at the `double`-precision floor for matrix eigenvalue computation on a well-conditioned 7×7 matrix — indistinguishable from exact.

4.6 Closed-form right-eigenvectors

The closed forms come from Stone & Gardiner 2008 Appendix B Eqs. (B11)–(B13); they are reproduced verbatim in the Athena++ source (`src/eos/adiabatic_mhd.cpp::LRMHDWaves`). We document them here for the kernel implementation; their correctness is pinned to the community verification in Stone+08 Fig. 28–30.

Define the **amplitude-normalisation coefficients**

$$\alpha_f^2 = \frac{c_{s_0}^2 - c_s^2}{c_f^2 - c_s^2}, \qquad \alpha_s^2 = \frac{c_f^2 - c_{s_0}^2}{c_f^2 - c_s^2}, \qquad \alpha_f^2 + \alpha_s^2 = 1,$$

and the **direction coefficients**

$$\beta_y = \frac{B_y}{\sqrt{B_y^2 + B_z^2}}, \qquad \beta_z = \frac{B_z}{\sqrt{B_y^2 + B_z^2}}, \qquad \beta_y^2 + \beta_z^2 = 1,$$

and $s = \mathrm{sign}(B_x)$.

4.6.1 Entropy (contact) wave — $\lambda_4 = v_x$

$$\mathbf{r}_{\text{entropy}} = (1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0)^{\mathrm{T}}.$$

Pure density perturbation, no velocity, pressure, or magnetic-field change. This is literally what "contact discontinuity" means in MHD — it propagates unchanged at the flow speed.

4.6.2 Alfvén waves — $\lambda_{2,6} = v_x \mp c_{Ax}$

$$\mathbf{r}_{A\pm} = \big(\,0,\ 0,\ \mp s\beta_z,\ \pm s\beta_y,\ -\beta_z\sqrt{\rho},\ \beta_y\sqrt{\rho},\ 0\,\big)^{\mathrm{T}}.$$

Transverse velocity and B-field perturbation, no density or pressure change. This is the canonical transverse electromagnetic wave of MHD.

4.6.3 Fast magnetosonic waves — $\lambda_{1,7} = v_x \mp c_f$

$$\mathbf{r}_{f\pm} = \Big(\,\rho\alpha_f,\ \pm\alpha_f c_f,\ \mp s\alpha_s c_s\beta_y,\ \mp s\alpha_s c_s\beta_z,\ \alpha_s c_{s_0}\sqrt{\rho}\beta_y,\ \alpha_s c_{s_0}\sqrt{\rho}\beta_z,\ \alpha_f\gamma p\,\Big)^{\mathrm{T}}.$$

Longitudinal compression plus in-phase perpendicular-B compression.

4.6.4 Slow magnetosonic waves — $\lambda_{3,5} = v_x \mp c_s$

$$\mathbf{r}_{s\pm} = \Big(\,\rho\alpha_s,\ \pm\alpha_s c_s,\ \pm s\alpha_f c_f\beta_y,\ \pm s\alpha_f c_f\beta_z,\ -\alpha_f c_{s_0}\sqrt{\rho}\beta_y,\ -\alpha_f c_{s_0}\sqrt{\rho}\beta_z,\ \alpha_s\gamma p\,\Big)^{\mathrm{T}}.$$

Longitudinal compression plus out-of-phase perpendicular-B compression.

4.7 Degenerate-limit eigenvectors

When $|\mathbf{B}_\perp| = 0$ (pure longitudinal $\mathbf{B}$), the β_y, β_z definitions above divide by zero. Stone+08 Eq. (B17)-(B20) gives replacement eigenvectors for this case. We do not verify these here — they are documented in the Athena++ source and must be implemented as branch-conditionals in the kernel.

4.8 □ Verification checkpoint (to be wired)

The future

`tests/test_athena_mhd_eigensystem.cu`

must:

1. For 100 random admissible states, evaluate `d_mhd_eigenvalues_primitive` and verify each returned λ_k agrees with (A3-wave-speeds) to 10^{-12} absolute.
2. For each state, evaluate `d_mhd_right_eigenvectors` and verify $\|\mathsf{A}_W(\mathbf{W})\,\mathbf{r}_k - \lambda_k \mathbf{r}_k\|_\infty < 10^{-12}$ for all $k \in \{1, \dots, 7\}$.
3. Pressure-positivity stress test: states with $\beta_{\text{plasma}} < 0.01$, verify no NaN / no overflow in α_f, α_s.
4. Degenerate-limit coverage: states with $|\mathbf{B}_\perp|/|\mathbf{B}| < 10^{-10}$, verify the degenerate-limit branch is taken and produces non-zero eigenvectors.

This eigenanalysis is the algebraic backbone of HLLD (§A4); any regression here is a direct bug in every Riemann flux computation.

5 A4. HLLD intermediate states (Miyoshi-Kusano 2005)

> **sympy script:** `scripts/a4_hlld_intermediate_states.py` **verified:** $p^\star_{\text{tot,L}} = p^\star_{\text{tot,R}}$ symbolically; Rankine–Hugoniot outer-wave jumps numerically (20 trials, max err 3×10^{-15}). **code checkpoints:** `athena_mhd_kernels.cu::d_hlld_flux`.

5.1 Wave fan

HLLD partitions the Riemann fan between S_L and S_R into four intermediate plateaus separated by $S_L^\star, S_M, S_R^\star$ — outer fast, Alfvén, contact, Alfvén, outer fast.

5.2 Contact speed (MK Eq. 38)

$$\boxed{S_M = \frac{(S_R - v_{xR})\rho_R v_{xR} - (S_L - v_{xL})\rho_L v_{xL} - p^\star_{\text{tot,R}} + p^\star_{\text{tot,L}}}{(S_R - v_{xR})\rho_R - (S_L - v_{xL})\rho_L}.}$$

5.3 Star-region state

$\rho_K^\star = \rho_K(S_K - v_{xK})/(S_K - S_M)$ (MK Eq. 43); $p^\star_{\text{tot}} = p^\star_{\text{tot,L}} + \rho_L(S_L - v_{xL})(S_M - v_{xL}) = p^\star_{\text{tot,R}} + \rho_R(S_R - v_{xR})(S_M - v_{xR})$ (MK Eq. 41). **Sympy symbolically verifies** the L/R equality.

$B^\star_{yK}, v^\star_{yK}$ from MK Eqs. 44-47 (see script). Alfvén speeds $S^\star_{L,R} = S_M \mp |B_x|/\sqrt{\rho^\star_{L,R}}$.

5.4 Critical implementation note

HLLD star state is NOT an EOS state. Do NOT compute $\mathbf{F}_K^\star$ via $p = (\gamma-1)(E^\star - \frac{1}{2}\rho|\mathbf{v}|^2 - \frac{1}{2}|\mathbf{B}|^2)$ — it gives the wrong pressure. Use MK Eq. 64 directly with $p^\star_{\text{tot}}$ from (A4-Ptot-star). Numerical verification confirms: RH residuals are $\mathcal{O}(1)$ with EOS inversion, $\mathcal{O}(10^{-15})$ with direct MK formula.

5.5 □ Verification

`tests/test_athena_mhd_hlld.cu` — Brio-Wu shock tube match Stone+08 Fig 28 to $L^1 < 2\%$ at $N = 512$.

6 A5. Constrained Transport and discrete $\nabla \cdot \mathbf{B} = 0$

sympy script: `scripts/a5_ct_divergence_preservation.py` **verified:** CT telescoping identity $(\nabla\cdot\mathbf{B})^{n+1} - (\nabla\cdot\mathbf{B})^n = 0$; Gardiner-Stone 2005 corner-EMF averaging is 2nd-order accurate. **code checkpoints:** `athena_mhd_solver.cu::update_face_B_with_emf`, `tests/test_athena_mhd_field_loop.cu`.

6.1 Grid layout

Yee-like staggered: B_x on vertical faces $(i \pm \frac{1}{2}, j)$, B_y on horizontal faces $(i, j \pm \frac{1}{2})$, E_z on corners.

6.2 CT update (Evans-Hawley 1988)

$$B_x^{i\pm1/2,j,n+1} = B_x^{i\pm1/2,j,n} - \frac{\Delta t}{\Delta y}(E_z^{i\pm1/2,j+1/2} - E_z^{i\pm1/2,j-1/2}),$$

$$B_y^{i,j\pm1/2,n+1} = B_y^{i,j\pm1/2,n} + \frac{\Delta t}{\Delta x}(E_z^{i+1/2,j\pm1/2} - E_z^{i-1/2,j\pm1/2}).$$

6.3 The central identity

$$\boxed{(\nabla \cdot \mathbf{B})_{i,j}^{n+1} = (\nabla \cdot \mathbf{B})_{i,j}^{n}}$$

Sympy symbolically verifies this as a pure telescoping identity — the four corner-EMF contributions cancel exactly, regardless of what values the EMFs take. This is why CT is qualitatively different from Dedner GLM: ∇·B = 0 is an **exact** property, not a truncation error.

6.4 Gardiner-Stone 2005 corner-EMF averaging

$$E_z^{i+1/2,j+1/2} = \tfrac{1}{4}(E_z^{x,i+1/2,j} + E_z^{x,i+1/2,j+1} + E_z^{y,i,j+1/2} + E_z^{y,i+1,j+1/2}).$$

Sympy-verified via Taylor expansion: leading error is $\frac{h^2}{8}(\partial_x^2 E_z + \partial_y^2 E_z) + \mathcal{O}(h^4)$, no $\mathcal{O}(h)$ term — matches 2nd-order Godunov accuracy.

6.5 □ Verification

`tests/test_athena_mhd_field_loop.cu` — GS05 field-loop, 10 crossings, lock max $|\nabla \cdot \mathbf{B}| < 10\varepsilon_{\text{mach}}$ at every step.

7 A6. Reconstruction (PLM/PPM) and TVD slope limiters

sympy script: `scripts/a6_reconstruction_tvd.py` **verified:** minmod / vL / MC opposite-sign cancellation; Sweby TVD region $0 \le \varphi(r) \le \min(2, 2r)$ for $r \in [0, 4]$; PLM reconstruction second-order; PPM 4-point interpolant 4th-order (exact to cubics); PPM parabola $\int_0^1 W\, d\xi = a_0$. **code checkpoints:** `athena_mhd_kernels.cu::d_reconstruct_primitive_plm`, `athena_mhd_kernels.cu::d_reconstruct_pri` `tests/test_athena_mhd_linear_wave_convergence.cu`.

7.1 Why this section matters

VL2 + HLLD without a **monotonic** reconstruction is 2nd-order but gives spurious oscillations at every shock. The three limiters below are the only ones we need in practice: **minmod** (safest, dispersive), **van Leer** (smooth, slightly sharper), **MC** (most aggressive, Stone+08 default).

A skipped step here directly causes the class of bugs already seen on stellar2d:

- `cart_ale2` swept-remap used donor-cell then MUSCL without characteristic projection — caused the periodic-BC drift at `P30/P31`.
- Stone+08 App. A is explicit: **reconstruct in primitive variables, then project onto characteristic variables** via the eigenvectors of §A3. Skipping the projection converts crisp fast-mode jumps into diffusive blobs.

7.2 Definitions

$$\sigma_{\text{minmod}}(a, b) = \begin{cases} 0, & \text{sign}(a) \neq \text{sign}(b), \\ \text{sign}(a)\ \min(|a|, |b|), & \text{else.} \end{cases}$$

$$\sigma_{\text{vL}}(a, b) = \begin{cases} 0, & ab \le 0, \\ \dfrac{2ab}{a+b}, & ab > 0. \end{cases}$$

$$\sigma_{\text{MC}}(a, b) = \text{sign}(a)\ \min\Big(2|a|, \tfrac{|a+b|}{2}, 2|b|\Big)\ \text{when } ab > 0,\ \text{else } 0.$$

7.3 Sweby TVD region (A6-Sweby)

All three limiters satisfy, with $r = b/a$:

$$\boxed{0 \le \varphi(r) \le \min(2,\, 2r), \quad r \ge 0.}$$

Sympy-verified by numerical sweep ($r \in [0, 4]$, 401 samples). Any limiter in this region is TVD for 1D scalar advection with CFL ≤ 1 (Harten 1983; LeVeque 2002 §16).

7.4 PLM reconstruction and order

$$W^L_{i+1/2} = W_i + \tfrac{1}{2}\sigma_i, \quad \sigma_i = \tfrac{1}{2}(W_{i+1} - W_{i-1}).$$

Leading error $W^L_{i+1/2} - W(x_{i+1/2}) = -\frac{h^2}{8}W''(x_{i+1/2}) + \mathcal{O}(h^3)$, so the reconstruction is **second order** (no $\mathcal{O}(h)$ term). Sympy-verified: no h^0 or h^1 terms in the residual expansion.

7.5 PPM (Colella-Woodward 1984) 4-point interpolant

$$\boxed{W_{i+1/2} = \tfrac{7}{12}(W_i + W_{i+1}) - \tfrac{1}{12}(W_{i-1} + W_{i+2}),}$$

applied to the **cell-averaged** values. Sympy-verified: no $\mathcal{O}(h^0..h^3)$ term in the expansion against the smooth reference; leading error is $-\frac{1}{30}h^4 W^{(4)}$.

The parabolic form inside a cell (Colella-Woodward 1984 Eq. 1.6),

$$W(\xi) = a_L + \xi[\Delta a + a_6(1-\xi)], \quad \Delta a = a_R - a_L, \quad a_6 = 6(a_0 - (a_L + a_R)/2),$$

satisfies $\int_0^1 W\, d\xi = a_0$ **exactly** (sympy-verified): conservation of cell averages is built in.

7.6 Practical recommendation (Stone+08 Appendix A)

1. Reconstruct $(\rho, \mathbf{v}, \mathbf{B}, p)$ in **primitive** variables (not conservative). This keeps positivity of ρ, p easier to enforce.
2. Project onto characteristic variables $\delta W^{(k)} = \ell_k \cdot \delta W$ using the left- eigenvectors of §A3, limit each wave family separately, then project back $\delta W = \sum_k r_k\, \delta W^{(k)}$.
3. Use MC as the default; switch to van Leer near strong shocks if MC produces staircase artefacts (rare but documented in Stone+08 Fig 28).

7.7 □ Verification checkpoints

- `tests/test_athena_mhd_linear_wave_convergence.cu` — 3-resolution linear fast-wave advection; L¹ convergence slope in $[1.9, 2.1]$.
- `tests/test_athena_mhd_shock_tube.cu` — Brio-Wu, MC limiter, no staircase within the fast rarefaction fan.

8 A7. Van-Leer 2 predictor-corrector (Stone-Gardiner 2009)

sympy script: `scripts/a7_vl2_predictor_corrector.py` **verified:** amplification factor $g(\xi) = 1 - i\nu\xi - \frac{\nu^2}{2}\xi^2 + \mathcal{O}(\xi^3)$ (Lax-Wendroff 2nd-order); $|g|^2 \le 1$ on $|\nu| \le 1$; truncation error $\mathcal{O}(\Delta t^3 + \Delta t\, \Delta x^2)$. **code checkpoints:** `athena_mhd_solver.cu::vl2_predictor_step`, `athena_mhd_solver.cu::vl2_corrector_step`.

8.1 The scheme

$$\boxed{\mathbf{U}^{\star} = \mathbf{U}^n + \tfrac{\Delta t}{2}\mathcal{L}[\mathbf{U}^n], \qquad \mathbf{U}^{n+1} = \mathbf{U}^n + \Delta t\,\mathcal{L}[\mathbf{U}^{\star}].}$$

This is **midpoint-RK2** in time, paired with any semi-discrete operator $\mathcal{L}$ (here: PLM reconstruction + HLLD flux from §A4, plus CT EMF update from §A5).

8.2 Why VL2 over Strang or CTU

Feature	VL2	Strang	CTU
Dimensional coupling	Unsplit, exact	Split, $\mathcal{O}(\Delta t^2)$ error	Unsplit
Auxiliary state	$\mathbf{U}^{\star}$	per-direction sweep	corner-transport cells
CFL limit (unsplit)	≤ 1	≤ 1 in each direction	≤ 1 (multi-D)
Compatibility with CT	Native (Stone-Gardiner 2009)	Requires extra corner EMF gymnastics	Native
Kernel complexity	Low (2 stages)	Low (per-direction)	Higher (CTU corner sweeps)

Stone+08 CTU is more accurate on discontinuities but significantly more complex; VL2 is the default in Athena++ and what we adopt.

8.3 Fourier (von Neumann) analysis on linear advection

With PLM + upwind flux for $a > 0$ on a smooth Fourier mode,

$$g(\xi) = 1 - i\nu\xi - \tfrac{\nu^2}{2}\xi^2 + \mathcal{O}(\xi^3), \quad \nu \equiv a\Delta t/\Delta x,$$

matching Lax-Wendroff at leading orders. Sympy-verified the three coefficients. Numerical sweep confirms $|g|^2 \leq 1$ for $\nu \in \{0.1, \ldots, 1.0\}$; at $\nu = 1.05$ the amplification is $|g|^2 \approx 1.22 > 1$, confirming the CFL limit is **sharp**.

8.4 Truncation error (A7-consistency)

With the central-flux semi-discrete operator $\mathcal{L}[U]_j = -a(U_{j+1} - U_{j-1})/(2h)$ acting on smooth U, after two VL2 stages,

$$\boxed{\mathbf{U}_j^{n+1} - U(x_j, t + \Delta t) = \mathcal{O}(\Delta t^3 + \Delta t\,\Delta x^2).}$$

Leading error is a purely dispersive $U_{xxx}\,a(a^2 - 1)/6$ term, vanishing at $\nu = 1$ (exact for advection on-CFL). Sympy expands the predictor/corrector composition and confirms the residual has no $\mathcal{O}(\varepsilon^0 \ldots \varepsilon^2)$ terms (with $\varepsilon = \Delta t = h$).

8.5 CFL constraint (1D)

$$\Delta t \le C_{\text{CFL}} \frac{\Delta x}{\max_{\text{cells}}(|v_x| + c_f)}, \quad C_{\text{CFL}} \le 1.$$

See §A8 for multidimensional generalisation and parabolic terms.

8.6 Implementation tips (for `athena_mhd_solver.cu`)

- Save $\mathbf{U}^n$ **before** the predictor so the corrector can rewind if a positivity check fails.
- The predictor need **not** compute the flux to full order — a 1st- order Godunov flux is enough; the corrector is where the 2nd-order accuracy is paid for. Stone+08 explicitly notes this economy.
- For CT, compute **face-centred** E_z at both predictor and corrector; average at the corners using Gardiner-Stone 2005 (§A5).

8.7 □ Verification checkpoints

- `tests/test_athena_mhd_linear_wave_convergence.cu` — three resolutions, fast wave, expect L¹ convergence slope in $[1.9, 2.1]$.
- `tests/test_athena_mhd_vl2_stability.cu` — CFL scan $\nu \in \{0.5, 0.9, 0.99\}$ stable; $\nu = 1.05$ must blow up.

9 A8. MHD CFL time-step constraints

sympy script: `scripts/a8_mhd_cfl.py` **verified:** FTCS diffusion amplification worst-case at $\xi = \pi$ gives $\sigma \le 1/2$; c_f limits in three degenerate cases ($B_\perp = 0$, $B_x = 0$, $\mathbf{B} = 0$). **code checkpoints:** `athena_mhd_kernels.cu::d_mhd_dt_reduction`, `athena_mhd_solver.cu::compute_dt`.

9.1 Hyperbolic CFL (fast wave dominant)

Combining §A3 (7-wave eigensystem) with §A7 ($|\nu| \le 1$), the unsplit multidimensional bound is

$$\boxed{\Delta t_{\text{hyp}} \le C_{\text{CFL}} \Bigg/ \sum_{d=1}^{D} \max_{\text{cells}} \left(\frac{|v_d| + c_{f,d}}{\Delta x_d} \right), \quad C_{\text{CFL}} \le 1,}$$

where the fast-magnetosonic speed $c_{f,d}$ in direction d is the positive root of the §A3 discriminant.

9.2 Parabolic CFL (Ohmic + ambipolar diffusion)

For the scalar diffusion $\partial_t U = \eta \partial_x^2 U$ with FTCS central space + forward Euler time, the amplification factor is $g(\xi) = 1 - 4\sigma \sin^2(\xi/2)$ with $\sigma = \eta \Delta t / \Delta x^2$. The worst case $\xi = \pi$ gives $g = 1 - 4\sigma$; stability $|g| \le 1$ requires $\sigma \le 1/2$ (sympy-verified).

Combining Ohmic + ambipolar as $\eta_{\text{eff}} = \eta_\Omega + \eta_{\text{AD}}$:

$$\boxed{\Delta t_{\text{para}} \le \tfrac{1}{2}\min_{\text{cells}} \frac{\Delta x^2}{\eta_\Omega + \eta_{\text{AD}}}.}$$

9.3 Fast-speed behaviour at degenerate limits (A8-cf-limits)

Sympy-verified:

Limit	c_f^2
$B_\perp = 0$ (tangential-free)	$\max(c_{s_0}^2, c_{Ax}^2)$
$B_x = 0$ (perpendicular only)	$c_{s_0}^2 + c_{A\perp}^2$
$\mathbf{B} = 0$ (hydro)	$c_{s_0}^2$

The first two use `numerical fall-back` in sympy because `sp.Max` does not simplify against the nested radical; 30 random states confirm agreement to machine precision.

9.4 Combined rule

$$\Delta t = \min\bigl(\Delta t_{\text{hyp}},\, \Delta t_{\text{para}}\bigr)\,.$$

9.5 Kernel-form reduction

Implementation-wise, per cell:

$$\left(\frac{1}{\Delta t}\right)_{i,j,k} = \frac{|v_x| + c_f}{\Delta x} + \frac{|v_y| + c_f}{\Delta y} + \frac{|v_z| + c_f}{\Delta z} + 2\frac{\eta_\Omega + \eta_{\text{AD}}}{\min(\Delta x, \Delta y, \Delta z)^2}.$$

Then Δt^{-1} is reduced (`max`) over the grid and the global Δt is $C_{\text{CFL}}/\Delta t^{-1}_{\text{max}}$.

9.6 Super-time-stepping (optional, future)

When η is large enough that $\Delta t_{\text{para}} \ll \Delta t_{\text{hyp}}$, the RKL2 super-time-stepping scheme (Meyer+12) advances the parabolic part with N sub-steps per hyperbolic step at $\mathcal{O}(N^2)$ CFL relaxation. Not included in the initial `athena_mhd` implementation; reserved for §C5 future extension when Suzuki turbulent heating is active.

9.7 □ Verification checkpoints

- `tests/test_athena_mhd_cfl_advection.cu` — advection of smooth wave at $C_{\text{CFL}} = 0.95$; no growth. At $C_{\text{CFL}} = 1.05$ the test **must** go unstable within 10 steps (positive control).
- `tests/test_athena_mhd_ohm_diffusion.cu` — pure Ohmic field-loop decay; Δt_{para} honored; analytic decay rate matched to $< 10^{-5}$.

10 A9. HLLD degenerate branches

sympy script: `scripts/a9_hlld_degeneracy.py` **verified:** on the Alfvén locus $(S_K - v_{xK})^2 = B_x^2/\rho_K$, numerator of $B_{yK}^\star$ vanishes and denominator reduces to $\rho_K(S_K - v_{xK})(v_{xK} - S_M)$; $D_{S_M} = 0$ when both sides of the Riemann interface are identical. **code checkpoints:** `athena_mhd_kernels.cu::d_hlld_flux` (branch dispatch).

10.1 Three degeneracies of the generic HLLD

The §A4 formulas $B_{yK}^\star, v_{yK}^\star$ blow up in three cases. The kernel must dispatch on these **before** evaluating the generic formulas.

Branch	Condition	Fallback
D1	$B_x = 0$	HLLC-MHD (Li 2005), 3-wave fan
D2	$(S_K - v_{xK})^2 = B_x^2/\rho_K$	Upstream B_y, v_y, B_z, v_z unchanged
D3	$S_L \to S_R$ and sides identical	Pure HLL (2-wave average)

10.2 D1 — Zero longitudinal field ($B_x = 0$)

$$B_x = 0 \Longrightarrow S_L^\star = S_R^\star = S_M, \ \mathbf{U}_K^\star = \mathbf{U}_K^{\star\star},$$

so the two-plateau star structure collapses to one plateau. The algorithm reduces to **HLLC-MHD** (Li 2005 Appendix A). Recommended implementation: detect $|B_x| < \varepsilon \cdot (c_{s_0} + c_{A\perp})\sqrt{\rho}$ and dispatch.

10.3 D2 — Alfvén locus removable singularity

The MK Eq. 44 denominator $\rho_K(S_K - v_{xK})(S_K - S_M) - B_x^2$ vanishes on the locus $(S_K - v_{xK})^2 = B_x^2/\rho_K$, where the upstream state already sits on the Alfvén characteristic.

Sympy-verified along the locus: - Numerator $\rho_K(S_K - v_{xK})^2 - B_x^2 \to 0$. - Denominator reduces to $\rho_K(S_K - v_{xK})(v_{xK} - S_M)$.

So both vanish, and $B_{yK}^\star$ is $0/0$ — *removable*. The physical content is that the transverse field does not jump across an Alfvén wave that coincides with the upstream state.

Kernel regularisation:

$$\boxed{|\text{den}| < \epsilon\sqrt{\rho_K}\,|B_x| \implies B_{yK}^\star \leftarrow B_{yK},\ v_{yK}^\star \leftarrow v_{yK} \text{ (and same for } z).}$$

with $\epsilon \sim 10^{-12}$ in double precision. Falling through to the upstream state is continuous with the generic formula.

10.4 D3 — $S_L \approx S_R$ (vacuum / strongly-aligned)

When $\|(S_R - v_{xR})\rho_R - (S_L - v_{xL})\rho_L\| < \epsilon$, the S_M denominator vanishes. Physically: the Riemann fan has collapsed (L = R), so there is no jump to resolve. Fall back to pure HLL:

$$\mathbf{F}_{\mathrm{HLL}} = \frac{S_R\mathbf{F}_L - S_L\mathbf{F}_R + S_L S_R(\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L}.$$

Pure HLL is diffusive but unconditionally stable, a safe last-resort.

10.5 Implementation order (most to least specific)

```
if |D_SM| < ε:
    return HLL_flux()                    # D3
if |B_x| < ε * (cs0 + cA⊥)*sqrt(ρ):
    return HLLC_MHD_flux()               # D1
compute generic star states (§A4)
if |den_K| < ε * sqrt(ρ_K) * |B_x|:      # D2
    set B_y*_K, B_z*_K, v_y*_K, v_z*_K
      to upstream values
return HLLD_flux(generic)
```

10.6 Why this matters

Without these dispatches, the Brio-Wu shock tube **will** produce NaN at $t \sim 0.05$ when a rarefaction fan grazes the Alfvén locus. Documented in Miyoshi-Kusano 2005 Sec. 3.4 and in Athena++ `src/hydro/rsolvers/mhd/hlld.cpp::HLLDTransport` branch logic.

10.7 □ Verification checkpoints

- `tests/test_athena_mhd_brio_wu.cu` — Brio-Wu, 512 cells, match Stone+08 Fig 28 to $L^1 < 2\%$.
- `tests/test_athena_mhd_degenerate_Bx0.cu` — synthetic state with $B_x = 10^{-14}$, must not produce NaN and must match pure-HLLC hydrodynamic flux to 10^{-10}.

11 A10. Powell 8-wave source vs Constrained Transport

sympy script: `scripts/a10_powell_vs_ct.py` **verified:** Powell source term $\mathbf{S}_{\mathrm{P}}$ vanishes identically when $\nabla \cdot \mathbf{B} = 0$; since §A5 gives CT preservation at the discrete level, $\mathbf{S}_{\mathrm{P}} \equiv 0$ for all n and no correction is needed. **code checkpoints:** no new code required — documented to prevent future "add a Powell source term for safety" PR.

11.1 The 8-wave system (Powell+99)

The 8-wave formulation augments ideal MHD with a divergence-cleaning wave and a **non-conservative** source term:

$$\partial_t\mathbf{U} + \partial_i\mathbf{F}_i(\mathbf{U}) = -(\nabla\cdot\mathbf{B})\,\mathbf{S}_{\mathrm{P}}(\mathbf{U}),$$

with

$$\mathbf{S}_{\mathrm{P}} = \begin{pmatrix} 0 \\ \mathbf{B} \\ \mathbf{v} \\ \mathbf{B}\cdot\mathbf{v} \end{pmatrix}.$$

At the **continuous** level, $\nabla \cdot \mathbf{B} = 0$ exactly, so $\mathbf{S}_{\mathrm{P}} \cdot 0 = 0$ — trivial.

11.2 The non-trivial claim

At the **discrete** level, a cell-centred or vertex-centred **B** storage generally has $(\nabla \cdot \mathbf{B})_{i,j} \neq 0$ at $\mathcal{O}(\Delta x)$, and the Powell source is a genuine correction. **But** on the Yee-staggered grid with CT (§A5), the discrete $(\nabla \cdot \mathbf{B})^n_{i,j}$ satisfies

$$(\nabla\cdot\mathbf{B})^n_{i,j} = (\nabla\cdot\mathbf{B})^0_{i,j} \; \forall\, n$$

(the §A5 telescoping identity). Provided initialisation seeds $\mathbf{B}^0$ from a vector potential or via one projection solve ($\nabla^2\phi = \nabla\cdot\mathbf{B}_{\mathrm{raw}}$, $\mathbf{B} \leftarrow \mathbf{B}_{\mathrm{raw}} - \nabla\phi$), $(\nabla\cdot\mathbf{B})^0 = 0$ and thus

$$\boxed{\mathbf{S}^n_{\mathrm{P}} \equiv 0 \; \forall n.}$$

11.3 GLM-MHD contrast (Dedner+02)

GLM adds an 8th field ψ and hyperbolic-parabolic cleaning:

$$\partial_t\psi + c_h^2(\nabla\cdot\mathbf{B}) = -\alpha\psi, \quad \partial_t\mathbf{B} + \nabla\psi = \ldots$$

GLM **does not** preserve $\nabla \cdot \mathbf{B} = 0$ exactly; it advects and damps the constraint violation at wave speed c_h. CT is *exact* by construction; GLM trades exactness for a grid that does not need face-centred field storage (ADER-DG codes etc.).

For `athena_mhd` we use CT. This section exists to lock in "no Powell source needed" as a *derived* result, not an assumption.

11.4 Implementation note

The kernel must:

1. Seed $\mathbf{B}^0$ from a vector potential **A** via $\mathbf{B} = \nabla \times \mathbf{A}$ at face centres. No projection needed; $\nabla \cdot \mathbf{B} \equiv 0$ by vector calculus.
2. Run the diagnostic `d_mhd_divB_check` after every time step to confirm $\max|\nabla \cdot \mathbf{B}| < 10\,\epsilon_{\mathrm{mach}}$. If this check ever fails, the bug is in Step 1 or in the CT kernel; not a reason to add a Powell source.

11.5 □ Verification checkpoints

- `tests/test_athena_mhd_field_loop.cu` — Gardiner-Stone 2005 field loop, 10 diagonal advections. Track $\max|\nabla \cdot \mathbf{B}|$ — must remain at machine precision.
- No test for the Powell source — it is **not** in the kernel and should never be added.

12 A11. Linear MHD wave initial conditions (Stone+08 Tab 1)

sympy script: `scripts/a11_linear_wave_initial_conditions.py` **verified:** with $\rho_0 = 1, p_0 = 1/\gamma, \mathbf{B}_0 = (1, \sqrt{2}, 1/2), \gamma = 5/3$ the wave speeds are $c_f = 2$,

$c_s = 1/2$, $c_{Ax} = 1$, $c_{s_0} = 1$, all matching Stone+08 Table 1; the discriminant identities $c_f^2 + c_s^2 = c_{s_0}^2 + c_A^2$ and $c_f^2 c_s^2 = c_{s_0}^2 c_{Ax}^2$ are sympy-verified. **code checkpoints:** `tests/test_athena_mhd_linear_wave_convergence.cu`; `tst/test_athena_mhd/test_linwave.py` (Python harness analogous to the existing `tst/test_ale2/test_linwave.py`).

12.1 Why this is the "gold-standard" convergence test

Linear wave advection has a known exact solution: after one period the perturbation returns to its initial shape, and any deviation is **purely numerical**. The L¹ norm of the difference, sampled at multiple resolutions, directly measures convergence slope and diagnoses accuracy loss of the scheme.

A scheme that claims 2nd order but gives slope 1.3 in the fast-wave test has a broken reconstruction, broken HLLD, or broken CT — and this test finds that before `athena_mhd` touches a shock tube.

12.2 Background (Stone+08 Table 1)

$$\rho_0 = 1, \quad p_0 = \frac{1}{\gamma}, \quad \mathbf{v}_0 = \mathbf{0}, \quad \mathbf{B}_0 = \left(1,\ \sqrt{2},\ \tfrac{1}{2}\right), \quad \gamma = \tfrac{5}{3}.$$

Derived wave speeds (all sympy-verified):

$$c_f = 2, \quad c_{Ax} = 1, \quad c_s = \tfrac{1}{2}, \quad c_{s_0} = 1.$$

12.3 Perturbation

For mode k with eigenvector $\mathbf{r}_k$ (from §A3) and amplitude $A = 10^{-6}$:

$$\mathbf{W}(x, 0) = \mathbf{W}_0 + A\,\mathbf{r}_k\ \cos\left(\frac{2\pi x}{L}\right), \quad x \in [0, L].$$

12.4 Periods on an L-periodic box

$$T_f = L/2, \quad T_A = L, \quad T_s = 2L.$$

12.5 Entropy-mode exception

The entropy eigenvalue is $\lambda_{\text{ent}} = v_{0,x}$ which is 0 at our base state. To make the entropy-wave convergence test meaningful, **choose** $v_{0,x} = 1$ **for the entropy-mode case only**; then $T_{\text{ent}} = L$. Stone+08 does the same (Sec 6.1).

12.6 $\nabla \cdot \mathbf{B} = 0$ at IC (automatic)

In the 1D MHD system (§A3) B_x is a parameter, not an evolution variable; all seven right-eigenvectors live in $(\rho, v_x, v_y, v_z, B_y, B_z, p)$ and do not perturb B_x. For a plane wave along $\hat{x}$, $\nabla \cdot \delta\mathbf{B} = ik_x \delta B_x = 0$ automatically. **No extra projection step needed at IC.**

12.7 Expected error

For a VL2 + HLLD + CT code (§A7, §A4, §A5),

$$\varepsilon_{L^1}(k, \Delta x) = C_k\, A\, (\Delta x/L)^2,$$

with C_k an $\mathcal{O}(1)$ mode-dependent constant. Convergence slope measured across $\Delta x = L/64, L/128, L/256$ should fall in $[1.9, 2.1]$ for all 7 modes.

12.8 Practical test protocol

1. For each of the 7 modes:
   - Run the solver at resolutions $N \in \{64, 128, 256\}$.
   - Evolve for one period T_k.
   - Compute $\varepsilon_{L^1}(N) = \frac{1}{N}\sum_j |W_j^N - W_j^{\text{exact}}|$.
2. Fit slope: $\log\varepsilon \propto p \log\Delta x$.
3. Accept if $|p-2| < 0.1$.

12.9 □ Verification checkpoint

`tests/test_athena_mhd_linear_wave_convergence.cu` — single test that cycles through 7 modes × 3 resolutions × 1 period; outputs a 21-row CSV; asserts slope ∈ [1.9, 2.1] per mode.

12.10 Extension: 3D linear wave

A 3D linear wave IC is constructed identically, substituting $k_x \to \mathbf{k}$ and rotating the eigenvector by the same angle. Stone+08 §6.2 does the oblique test at $\mathbf{k} = (1, 2, 2)$. This stresses dimensional coupling. Reserved for post-MVP; not included in the initial test suite.

13 B1. Super-radial flux-tube reduction

sympy script: `scripts/b1_flux_tube_geometry.py` **verified:** flux conservation $AB_r = \text{const}$; MHSE $\partial_r p + \rho g = -B_r^2 \partial_r(\ln A)$; Kopp-Holzer $f(R_*) = 1$, $f(\infty) = f_{\max}$. **code checkpoints:** `tests/test_mhd_wind_hse_stationary.cu`.

13.1 Geometry

$A(r) = r^2 f(r)$, $f(R_*) = 1$, $f(\infty) = f_{\max}$. Kopp-Holzer form:

$$f(r) = \frac{f_{\max}\, e^{(r-R_1)/\sigma} + f_1}{e^{(r-R_1)/\sigma} + 1}, \quad f_1 = 1 - (f_{\max} - 1) e^{(R_* - R_1)/\sigma}.$$

13.2 1D flux-tube equations

$$\partial_t(\rho A) + \partial_r(\rho v_r A) = 0,$$
$$\partial_t(\rho v_r A) + \partial_r[A(\rho v_r^2 + p_{\text{tot}} - B_r^2)] = (p_{\text{tot}} - B_r^2)\partial_r A - \rho g A,$$
$$\partial_t(EA) + \partial_r[A((E + p_{\text{tot}})v_r - B_r(\mathbf{B}\cdot\mathbf{v}))] = -\rho g v_r A + AQ.$$

13.3 Flux conservation

$$\boxed{A(r)B_r(r) = R_*^2 B_0 \Rightarrow B_r(r) = \frac{B_0}{f(r)}(R_*/r)^2.}$$

13.4 MHSE

$$\boxed{\partial_r p + \rho g = -B_r^2\,\partial_r(\ln A).}$$

IC gotcha: for a Suzuki-style wind run, atmospheres must be integrated from this MHSE, NOT plain HSE. Using HSE triggers a $\sim 10^{-2}$ transient and corrupts $\dot{M}$ measurements.

13.5 WKB amplitude (see B2)

$\delta v_\perp \propto (\rho v_A A)^{-1/2}$ (Poynting) or $(\rho v_A A)^{-1/4}$ (per-mode amplitude convention).

13.6 □ Verification

`tests/test_mhd_wind_hse_stationary.cu` — HSE atmosphere, 100 acoustic crossings, lock max $|v_r|/c_s < 10^{-4}$.

14 B2. WKB wave action conservation for Alfvén waves

> **sympy script:** `scripts/b2_wave_action_wkb.py` **verified:** dispersion $\omega^2 = v_A^2 k^2$; no-reflection limit under uniform $\rho_0, B_{r,0}$. **code checkpoints:** `athena_mhd_kernels.cu::d_alfven_wave` `tests/test_mhd_wind_amplitude_scaling.cu`.

14.1 Linearised transverse MHD

Around a static background $(\rho_0(r), B_{r,0}(r))$ with $v_{r,0} = 0$:

$$\partial_t v_\perp = \frac{B_{r,0}}{\rho_0}\partial_r B_\perp, \qquad \partial_t B_\perp = B_{r,0}\partial_r v_\perp. \quad \text{(B2-linear)}$$

WKB ansatz gives the Alfvén dispersion

$$\boxed{\omega^2 = v_A^2 k^2, \qquad v_A \equiv B_{r,0}/\sqrt{\rho_0}.} \quad \text{(B2-disp)}$$

Sympy verified that the determinant of the 2×2 linearised matrix equals $-\omega^2 + v_A^2 k^2$, giving the dispersion as the unique non-trivial mode.

14.2 Elsässer variables

$z_\pm \equiv v_\perp \mp B_\perp/\sqrt{\rho_0}$ decouple the outgoing and incoming Alfvén waves:

$$\partial_t z_\pm \pm v_A\,\partial_r z_\pm = S_{\text{refl}}\,z_\mp, \quad \text{(B2-Elsasser)}$$

with a reflection source S_{refl} that depends on the background gradients. In a uniform $(\rho_0, B_{r,0})$, sympy verifies $S_{\text{refl}} = 0$ — no wave reflection in a uniform atmosphere (Alfvén waves are pure one-way in the uniform limit).

14.3 Wave action conservation

Outgoing z_+ in the absence of reflection conserves the wave action

$$\boxed{\rho_0 A\,|z_+|^2/v_A = \text{const along characteristics.}} \quad \text{(B2-action)}$$

For the amplitude-scaling law of a driver BC (relevant for Suzuki winds), **Poynting flux** $\rho_0 v_A A|\delta v_\perp|^2$ conservation gives

$$|\delta v_\perp| \propto (\rho_0 v_A A)^{-1/2}, \quad \text{(B2-Poynting)}$$

while a **per-mode** amplitude normalisation gives the Jacques-1977 form

$$|\delta v_\perp| \propto (\rho_0 v_A A)^{-1/4}. \quad \text{(B2-Jacques)}$$

The two differ because the former fixes the transmitted energy flux while the latter fixes the wave amplitude in Elsässer space. Suzuki wind codes use **the Poynting-flux convention** at the photospheric driver (amplitude $\langle \delta v_\perp \rangle \approx 1.25$ km/s in Shimizu+22) — make sure the kernel matches.

14.4 □ Verification

`tests/test_mhd_wind_amplitude_scaling.cu` — linearised Alfvén wave from a fixed BC driver. Measure $\delta v_\perp^{\text{rms}}(r)$ at $r = 5, 10, 20\,R_*$; lock deviation from $(\rho v_A A)^{-1/2}$ at $< 5\%$.

15 B3. Parker critical point on a super-radial flux tube

> **sympy script:** `scripts/b3_parker_critical_point.py` **verified:** Parker wind equation derivation; classical spherical limit $r_c = GM_*/(2c_s^2)$. **code checkpoints:** `tests/test_mhd_parker_wind.cu`.

15.1 Parker wind equation

Steady isothermal flow on a super-radial tube $(A(r) = r^2 f(r))$ with $p = c_s^2 \rho$ yields, after eliminating ρ via mass conservation:

$$\boxed{\left(v - \frac{c_s^2}{v}\right)\frac{dv}{dr} = c_s^2\,\frac{d\ln A}{dr} - \frac{GM_*}{r^2}.} \quad \text{(B3-Parker)}$$

Sympy derived from mass + momentum + EOS.

15.2 Critical (sonic) point

At $v = c_s$ the LHS vanishes, forcing

$$c_s^2 \left(\frac{2}{r_c} + \left. \frac{d \ln f}{dr} \right|_{r_c} \right) = \frac{GM_*}{r_c^2}. \quad \text{(B3-critical)}$$

Spherical limit $f \equiv 1$: $r_c = GM_*/(2c_s^2)$ (classical Parker radius). Sympy verifies this by solving (B3-critical) analytically.

15.3 Asymptotic velocity

Far from r_c (spherical limit), $v(r) \sim c_s \sqrt{4 \ln(r/r_c) + \text{const}}$ — logarithmic growth, characteristic of the Parker isothermal wind.

15.4 □ Verification

`tests/test_mhd_parker_wind.cu` — isothermal, no magnetic driver, set $T = 2 \times 10^6$ K, $M_* = 1 M_\odot$, measure Mach number crossing at $r = r_c$. Lock $|M(r_c) - 1| < 10^{-3}$.

16 B4. Well-balanced MHSE at the operator level

sympy script: `scripts/b4_well_balanced_mhse.py` **verified:** $F_{\text{wb}}(\mathbf{U}_{\text{hse}}) \equiv 0$ by construction; linear-wave Jacobian preserved, $\partial_{\mathbf{U}} F_{\text{wb}} = \partial_{\mathbf{U}} R$ at $\mathbf{U}_{\text{hse}}$ — perturbation dynamics unchanged. **code checkpoints:** `athena_mhd_solver.cu::compute_residual_wb`, `athena_mhd_solver.cu::snapshot_hse_if_needed`, `tests/test_athena_mhd_wind_hse_stationary.cu`.

16.1 The problem

Direct application of the discrete residual $R(\mathbf{U}) = -\partial_i \mathbf{F}_i + \mathbf{S}$ to a piecewise- constant MHSE atmosphere leaves a truncation residual of order $\mathcal{O}(\Delta r^2)\, \rho g$. For a Suzuki-style wind run this residual drives a $|\delta v_r|/c_s \sim 10^{-2}$ transient that corrupts the wind-mass-loss diagnostic for hundreds of crossing times.

16.2 Well-balanced residual (Bermúdez-Vázquez 1994 / Botta+04)

$$\boxed{F_{\text{wb}}(\mathbf{U}) \equiv R(\mathbf{U}) - R(\mathbf{U}_{\text{hse}}).}$$

By construction $F_{\text{wb}}(\mathbf{U}_{\text{hse}}) = 0$ (sympy-verified trivially). So an atmosphere at MHSE stays at MHSE to machine precision for arbitrarily long time, regardless of reconstruction order or Riemann solver.

16.3 Preservation of linear-wave dynamics

A small perturbation $\delta \mathbf{U}$ around the MHSE state:

$$F_{\text{wb}}(\mathbf{U}_{\text{hse}} + \delta \mathbf{U}) = \left. \partial_{\mathbf{U}} R \right|_{\mathbf{U}_{\text{hse}}} \cdot \delta \mathbf{U} + \mathcal{O}(\delta \mathbf{U}^2).$$

Sympy-verified that the Jacobian of F_{wb} at $\mathbf{U}_{\text{hse}}$ **equals** the Jacobian of R at the same state. Thus linear MHD waves (Alfvén, magnetosonic) propagate through F_{wb} identically to R — no spurious damping or dispersion is introduced by the well-balancing subtraction.

16.4 What MHSE means in a super-radial flux tube

Reprise from §B1:

$$\boxed{\partial_r p + \rho g + B_r^2\,\partial_r(\ln A) = 0.}$$

Note the **last term** is critical. A naïve WB that only cancels gravity (the hydrodynamic well-balancing) will leave an $\mathcal{O}(B_r^2)$ residual. The correction for the super-radial tube must include the area-divergence term.

16.5 Practical recipe

1. **Snapshot MHSE on the grid.** Integrate the ODE $\partial_r p_{\text{hse}} = -\rho_{\text{hse}} g - B_r^2\,\partial_r(\ln A)$ using the given $(\rho_{\text{hse}}, B_r, A)$ profiles.
2. **Compute** $R(\mathbf{U}_{\text{hse}})$ **once** at startup, store per cell. (Assumes $\mathbf{U}_{\text{hse}}$ is time-independent — true for stationary background.)
3. **Evolve** $\partial_t \mathbf{U} = R(\mathbf{U}) - R(\mathbf{U}_{\text{hse}})$.
4. **Initial condition**: seed $\mathbf{U}^0 = \mathbf{U}_{\text{hse}}$. Initial RHS identically zero → no transient.

16.6 Failure modes observed elsewhere

- `radial1d` without this correction: Newton iteration looks converged ($|F| < 10^{-9}$) but the HSE slowly drifts at machine precision × resolution. Seen in pre-MS KH attempts: HSE-Newton is stable but cannot initiate KH because $F_v \equiv 0$ at MHSE.
- `cart_ale2` with WB: integration-grade HSE stability for 10^4 crossings.

16.7 Snapshot invalidation

If $(\rho_{\text{hse}}, B_r, A)$ profiles change (e.g., user-driven parameter sweep), re-snapshot. The kernel can detect staleness via a hash of the profile arrays.

16.8 □ Verification checkpoints

- `tests/test_athena_mhd_wind_hse_stationary.cu` — MHSE atmosphere, 10^4 acoustic crossings, assert max $|v_r|/c_s < 10^{-8}$ (with WB on) vs $\sim 10^{-2}$ (with WB off). The "off" case is a **positive control**: if turning off WB doesn't produce a transient, the WB machinery is a no-op and should be audited.

17 C1. Ohmic (resistive) dissipation

sympy script: `scripts/c1_ohmic_dissipation.py` **verified:** $\nabla \times (\nabla \times B) = \nabla(\nabla\cdot B) - \nabla^2 B$ (all 3 components); $\nabla\times(\eta_O J) = \eta_O \nabla\times J + (\nabla\eta_O)\times J$. **code checkpoints:** `athena_mhd_kernels.cu::d_ohmic_dissipation`, `tests/test_mhd_ohmic_diffusion.cu`.

17.1 Ohm's law with finite conductivity

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} + \eta_O \mathbf{J}, \qquad \mathbf{J} = \nabla \times \mathbf{B}. \quad \text{(C1-Ohm)}$$

17.2 Induction equation

Constant η_O:

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta_O \nabla^2 \mathbf{B}. \quad \text{(C1-induction-const)}$$

Variable $\eta_O(\mathbf{r})$:

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta_O \nabla^2 \mathbf{B} - (\nabla \eta_O) \times \mathbf{J}. \quad \text{(C1-induction-var)}$$

Sympy verified via the vector identity $\nabla \times (\eta_O \mathbf{J}) = \eta_O \nabla \times \mathbf{J} + (\nabla \eta_O) \times \mathbf{J}$.

17.3 Joule heating

$$\boxed{Q_{\text{Ohm}} = \eta_O |\mathbf{J}|^2 \geq 0.} \quad \text{(C1-Q)}$$

Manifestly non-negative; enters internal-energy source.

17.4 CFL for explicit diffusion

$$\Delta t \leq \frac{(\Delta x)^2}{2\eta_O} \text{ (1D)}, \ \frac{(\Delta x)^2}{4\eta_O} \text{ (2D)}, \ \frac{(\Delta x)^2}{6\eta_O} \text{ (3D)}. \quad \text{(C1-CFL)}$$

At $r \approx 1000$ km in Matsuoka+24, $\eta_O \sim 10^{12}$ cm²/s and $\Delta x \sim 1$ km, so $\Delta t_{\text{Ohm}} \sim 10^{-4}$ s — much smaller than hydro CFL. In practice, Matsuoka+24 use **super- time-stepping** (Alexiades-Amiez-Gremaud 1996) to accelerate the explicit Ohmic update.

17.5 ▯ Verification

`tests/test_mhd_ohmic_diffusion.cu` — sinusoidal B_y perturbation, $▯_O = $ const. Lock $L^2(B_y)$ exponential decay rate matches analytical $e^{-\eta_O k^2 t}$ to $< 1\%$ at $N = 128$.

18 C2. Ambipolar diffusion (ion-neutral drift)

> **sympy script:** `scripts/c2_ambipolar_dissipation.py` **verified:** $\mathbf{J}_\perp \cdot \hat{\mathbf{B}} = 0$; $\mathbf{J}_\parallel \times \mathbf{B} = 0$; $(\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}} = -\mathbf{J}_\perp$; $(\mathbf{J} \times \mathbf{B}) \times \mathbf{B}/|\mathbf{B}|^2 = (\mathbf{J} \times \hat{\mathbf{B}}) \times \hat{\mathbf{B}}$; $Q_{\text{amb}} = \eta_A |\mathbf{J}_\perp|^2 \geq 0$. **code checkpoints:** `athena_mhd_kernels.cu::d_ambipolar_flux`, `tests/test_mhd_ambipolar_bmin.cu`.

18.1 Motivation

In a partially-ionised plasma, **neutrals** do not couple directly to **B**; they feel the Lorentz force only via ion-neutral collisions. When the collision rate ν_{in} is slower than MHD timescales, ions drift relative to neutrals at

$$\mathbf{v}_{\text{drift}} = \mathbf{J} \times \mathbf{B}/(\rho_i \rho_n \nu_{in}),$$

yielding the non-ideal electric field

$$\mathbf{E}_{\text{amb}} = \eta_A\,(\mathbf{J}\times\hat{\mathbf{B}})\times\hat{\mathbf{B}} = -\eta_A\,\mathbf{J}_\perp, \quad \text{(C2-Eamb)}$$

with $\eta_A \equiv |\mathbf{B}|^2/(\rho_i\rho_n\gamma_{in})$.

18.2 Parallel / perpendicular current selectivity

Critical property: only $\mathbf{J}_\perp$ is dissipated by ambipolar; the parallel current $\mathbf{J}_\parallel$ flows freely along $\hat{\mathbf{B}}$:

$$\mathbf{J}_\parallel \times \mathbf{B} = \mathbf{0}. \quad \text{(C2-selective)}$$

18.3 Kernel-friendly form

For implementation, the non-unit-vector form is preferred:

$$\boxed{\mathbf{E}_{\text{amb}} = \eta_A\,\frac{(\mathbf{J}\times\mathbf{B})\times\mathbf{B}}{|\mathbf{B}|^2},} \quad \text{(C2-altform)}$$

avoids dividing by $|\mathbf{B}|$ in the normalisation step — matters for low-$|\mathbf{B}|$ robustness (chromosphere bottoms).

18.4 Induction equation

$$\partial_t\mathbf{B} = \nabla\times(\mathbf{v}\times\mathbf{B}) + \nabla\times\left[\eta_A\frac{(\mathbf{J}\times\mathbf{B})\times\mathbf{B}}{|\mathbf{B}|^2}\right]. \quad \text{(C2-induction)}$$

Non-linear in B — key difference from Ohmic.

18.5 Ambipolar heating

$$\boxed{Q_{\text{amb}} = \eta_A|\mathbf{J}_\perp|^2 = \eta_A\frac{|\mathbf{J}|^2|\mathbf{B}|^2 - (\mathbf{J}\cdot\mathbf{B})^2}{|\mathbf{B}|^2} \geq 0.} \quad \text{(C2-Q)}$$

18.6 CFL

$$\Delta t \leq \frac{(\Delta x)^2}{2\eta_A|\mathbf{B}|^2/\rho}\ (perdirection). \quad \text{(C2-CFL)}$$

Scales with $|\mathbf{B}|^2$ — tight CFL in strong-field regions.

18.7 □ Verification

`tests/test_mhd_ambipolar_bmin.cu` — B_x uniform, sinusoidal B_y with \$□_A = \$ const. Lock decay rate against analytical; also verify that when $B_y \parallel B_x$ (i.e., $\mathbf{J}\parallel\mathbf{B}$), $Q_{\text{amb}} = 0$ and B_y does not decay. Catches any accidental scalar-diffusion substitution for the tensor form.

19 C3. Total-energy equation with non-ideal MHD

sympy script: `scripts/c3_resistive_energy.py` **verified:** ideal Poynting flux identity; $Q_{\mathrm{Ohm}} = \eta_O |J|^2$; $Q_{\mathrm{amb}} = \eta_A |J_\perp|^2$ (via Lagrange's identity). **code checkpoints:** `athena_mhd_kernels.cu::d_total_energy_flux`.

19.1 Total-energy equation

With both Ohmic + ambipolar, total energy is still conserved; only the Poynting flux gets a non-ideal addition:

$$\boxed{\partial_t E + \nabla \cdot [(E + p^\star)\mathbf{v} - \mathbf{B}(\mathbf{v} \cdot \mathbf{B}) + \mathbf{E}_{\mathrm{ni}} \times \mathbf{B}] = 0,} \quad \text{(C3-energy)}$$

with $\mathbf{E}_{\mathrm{ni}} = \eta_O \mathbf{J} + \eta_A (\mathbf{J} \times \mathbf{B}) \times \mathbf{B}/|\mathbf{B}|^2$.

19.2 Internal-energy source

The non-ideal dissipation becomes internal-energy heating:

$$\boxed{\partial_t(\rho e) + \nabla \cdot (\rho e\, \mathbf{v}) = -p \nabla \cdot \mathbf{v} + Q_{\mathrm{Ohm}} + Q_{\mathrm{amb}},} \quad \text{(C3-internal)}$$

$$Q_{\mathrm{Ohm}} = \eta_O |\mathbf{J}|^2, \qquad Q_{\mathrm{amb}} = \eta_A |\mathbf{J}_\perp|^2 = \eta_A \frac{|\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2}{|\mathbf{B}|^2}.$$

Sympy verified the Lagrange identity $|\mathbf{J} \times \mathbf{B}|^2 = |\mathbf{J}|^2 |\mathbf{B}|^2 - (\mathbf{J} \cdot \mathbf{B})^2$ which maps between the two Q_amb forms.

19.3 Sign convention

$Q = -\mathbf{E}_{\mathrm{ni}} \cdot \mathbf{J}$ is the rate at which EM energy is dissipated **into the fluid's internal energy** — positive for dissipative $\mathbf{E}_{\mathrm{ni}}$. Both Ohmic and ambipolar give $Q > 0$.

19.4 Kernel structure

- **Total energy**: evolved via conservative flux (C3-energy). Non-ideal Poynting flux $\mathbf{E}_{\mathrm{ni}} \times \mathbf{B}$ added to the hydro energy flux.
- **No direct Q kernel needed**: internal-energy source is implicitly accounted for by the flux divergence of the non-ideal Poynting term. Explicit computation of Q only needed for diagnostics.

19.5 □ Verification

`tests/test_mhd_ohmic_energy_conservation.cu` — uniform $\mathbf{B}_0$ with sinusoidal B_y perturbation + $□_O = $ const. Measure total energy drift over 100 decay timescales. Lock $|\Delta E/E_0| < 10^{-8}$ — **any leak means the non-ideal Poynting flux term is missing from the energy flux**.

20 C4. Saha closure for η_O and η_A

sympy script: `scripts/c4_saha_ionization_closure.py` **verified:** Saha low-T limit $x_e \to 0$; high-T limit $x_e \to 1$. **code checkpoints:** `athena_mhd_kernels.cu::d_saha_eta`, `tests/test_mhd_saha_table.cu`.

20.1 Saha equation (pure hydrogen, LTE)

$$\boxed{\frac{x_e^2}{1-x_e} = \frac{1}{n_H}\left(\frac{m_e k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_H/k_B T},} \quad \text{(C4-Saha)}$$

with $\chi_H = 13.6$ eV and $x_e = n_e/(n_e + n_H)$. **Sympy verified** $x_e \to 0$ as $T \to 0$ and $x_e \to 1$ as $T \to \infty$.

20.2 Diffusivity closures (Draine 1983 / Choi+09)

Ohmic (electron-neutral collisions):

$$\eta_O \approx 234\sqrt{T/10^4\,\mathrm{K}}\, x_e^{-1}\ \mathrm{cm}^2/\mathrm{s}. \quad \text{(C4-etaO)}$$

Ambipolar (ion-neutral drift):

$$\eta_A = \frac{|\mathbf{B}|^2}{\rho_i \rho_n \gamma_{in}}, \qquad \gamma_{in} \approx 3.5 \times 10^{13}\ \mathrm{cm}^3/\mathrm{g/s}. \quad \text{(C4-etaA)}$$

These are the forms used in Suzuki+25 (RGB winds) and Matsuoka+24 (solar chromosphere).

20.3 Weakly-ionised limit ($x_e \ll 1$)

$$\eta_O \sim \frac{234\sqrt{T/10^4}}{x_e}, \qquad \eta_A \sim \frac{|\mathbf{B}|^2}{x_e \rho^2 \gamma_{in}}.$$

Both diverge as $x_e \to 0$ — non-ideal effects are maximal at the chromosphere base where most hydrogen is neutral. This is why the RGB winds in Suzuki+25 get their 15× $\dot{M}$ suppression from ambipolar: it shuts down high-frequency Alfvén wave transmission at the chromospheric floor.

20.4 Magnetic Reynolds numbers

$$R_m^{\mathrm{Ohm}} = \frac{L v_A}{\eta_O}, \qquad R_m^{\mathrm{amb}} = \frac{L v_A \rho}{\eta_A |\mathbf{B}|^2}.$$

Ideal MHD requires $R_m \gg 1$**.** Matsuoka+24 reports $R_m \sim 1 - 10$ at $r \approx 1000$ km in the solar chromosphere — non-ideal MHD kicks in below this altitude.

20.5 Kernel implementation

Pre-tabulate (η_O, η_A) as functions of $(\rho, T, |\mathbf{B}|)$ on a 3D table to avoid evaluating the Saha equation every timestep. Suzuki+25 uses a 128×128×64 log-spaced table; cost is negligible once built.

20.6 □ Verification

`tests/test_mhd_saha_table.cu` — verify tabulated (η_O, η_A) match Draine 1983 Table 2 to $< 1\%$ at five reference (T, ρ) pairs: $(10^4, 10^{-9})$, $(6000, 10^{-7})$, $(4500, 10^{-8})$, $(3000, 10^{-11})$, $(10^5, 10^{-12})$.

21 C5. Sub-grid turbulent heating closure (Suzuki-Inutsuka 2005)

> **sympy script:** `scripts/c5_suzuki_turbulent_heating.py` **verified:** positivity $\varepsilon_{\text{turb}} > 0$; pure-outward limit $|z^-| \to 0$ gives no cascade ($\varepsilon^{\text{SY}}_{\text{turb}} \to 0$); timescale bound $\Delta t \le \lambda_{\text{cor}}/(2c_d|\delta v_\perp|)$. **code checkpoints:** `athena_mhd_kernels.cu::d_turbulent_heating_sour` `tests/test_athena_mhd_turbulent_heating_positivity.cu`.

21.1 Why this closure exists

The 1D super-radial flux-tube code (§B1) cannot resolve the 3D Alfvén wave turbulence that transfers outward-going wave energy to heat in the corona. Suzuki-Inutsuka 2005 (SI05) added a phenomenological sub-grid heating term that dissipates transverse wave energy at a prescribed cascade rate. This is the crucial ingredient without which the 1D wind models severely **under-estimate** coronal temperature.

21.2 Suzuki-Inutsuka 2005 form

$$\boxed{\varepsilon^{\text{SI}}_{\text{turb}} = c_d\,\rho\,\frac{|\delta v_\perp|^3}{\lambda_{\text{cor}}}, \quad c_d \approx 0.1, \quad \lambda_{\text{cor}} \sim 10^7\,\text{cm}.}$$

Dimensions: $[\rho][\delta v]^3/[\lambda] = \text{erg}/\text{cm}^3/\text{s}$ □.

21.3 Shoda-Yokoyama 2016 (Elsässer) form

$$\varepsilon^{\text{SY}}_{\text{turb}} = \frac{\rho}{2\lambda_{\text{cor}}}\left(|\mathbf{z}^+|^2|\mathbf{z}^-| + |\mathbf{z}^-|^2|\mathbf{z}^+|\right).$$

This form makes the **non-linear coupling** between counter- propagating modes explicit. Limit $|z^-| \to 0$ gives $\varepsilon^{\text{SY}}_{\text{turb}} \to 0$ (sympy-verified): no cascade without cross-polarisation — an essential physical consistency check.

21.4 Energy-equation coupling

$$\partial_t E + \nabla\cdot[(E + P^\star)\mathbf{v} - \mathbf{B}(\mathbf{B}\cdot\mathbf{v})] = \varepsilon_{\text{turb}} \ \ge 0.$$

Positivity. Sympy-verified $\varepsilon_{\text{turb}} > 0$ for positive $\rho, \delta v_\perp, \lambda_{\text{cor}}, c_d$. The kernel must not inadvertently create a negative-definite heating source — any code review must check this.

21.5 Timescale bound (explicit-source CFL)

Per-step heating must not exceed the available wave KE:

$$\varepsilon_{\text{turb}} \cdot \Delta t \le \tfrac{1}{2}\rho|\delta v_\perp|^2 \iff \Delta t \le \frac{\lambda_{\text{cor}}}{2\, c_d\, |\delta v_\perp|}.$$

In practice, for solar wind parameters ($|\delta v_\perp| \sim 30\,\text{km/s}$, $\lambda_{\text{cor}} \sim 10^8\,\text{cm}$), this bound is well above the hyperbolic CFL; no concern.

21.6 Shimizu+22 scaling

$$\lambda_{\text{cor}}(r) = \lambda_0 \sqrt{A(r)/A(R_*)}, \quad \lambda_0 \approx 10^7\,\text{cm}$$

(i.e., correlation length scales with the tube area because energy- containing eddies fill the tube cross-section). Replication of Shimizu+22 figures requires this scaling **and** $c_d = 0.1$ exactly — the paper's calibration.

21.7 Extraction of $\delta v_\perp$

$\delta v_\perp$ is defined in the **Lagrangian / comoving frame**:

$$|\delta v_\perp|^2 \equiv \langle (v_y - \bar{v}_y)^2 + (v_z - \bar{v}_z)^2 \rangle,$$

with $\bar{v}$ a running time-average. In a 1D tube code with no transverse degree of freedom, this can instead be derived from the **Alfvén wave amplitude**

$$|\delta v_\perp|^2 = |z^+|^2 + |z^-|^2 - 2\mathbf{z}^+ \cdot \mathbf{z}^-$$

using the Elsässer variables tracked directly from §B2.

21.8 □ Verification checkpoints

- `tests/test_athena_mhd_turbulent_heating_positivity.cu` — random initial $(\rho, \delta v, \lambda)$ states, assert $\varepsilon_{\text{turb}} \ge 0$ across 100 samples; no NaN.
- `tests/test_athena_mhd_suzuki_wind.cu` — full replication of SI05 Fig. 2 mass-loss rate $\dot{M} \sim 2 \times 10^{-14}\, M_\odot/\text{yr}$ within ±10% (calibration tolerance per paper's own sensitivity to c_d). Requires well-balanced MHSE (§B4).

22 C6. Spitzer-Härm anisotropic thermal conduction

sympy script: `scripts/c6_spitzer_conduction.py` **verified:** parallel projector $\mathsf{P}_\parallel = \hat{\mathbf{b}}\hat{\mathbf{b}}^{\mathrm{T}}$ idempotent with $\operatorname{tr}\mathsf{P}_\parallel = 1$; parallel / perpendicular decomposition of ∇T; Kirchhoff potential identity $\mathbf{F}_c = -\nabla[\frac{2}{7}\kappa_0 T^{7/2}]$ in 1D; entropy production $\sigma_{\text{cond}} = \kappa_\parallel(\hat{\mathbf{b}} \cdot \nabla T)^2/T^2 \ge 0$; FTCS stability $\sigma \le 1/2$. **code checkpoints:** `athena_mhd_kernels.cu::d_spitzer_heat_flux`, `athena_mhd_solver.cu::compute_conduction_dt`, `tests/test_athena_mhd_conduction_isothermal.cu`.

22.1 Why Spitzer conduction dominates the corona

In a fully-ionised, magnetised plasma the electron mean-free-path scales as $\lambda_e \propto T^2/n_e$, while the transit time between collisions $\tau_c \propto T^{3/2}/n_e$. Combining these with the electron thermal velocity $v_{th,e} \propto \sqrt{T}$ gives the Spitzer-Härm (1953) / Braginskii (1965) conductivity

$$\kappa_\parallel(T) = \kappa_0\, T^{5/2}, \quad \kappa_0 \approx 10^{-6}\ \mathrm{erg\,cm^{-1}\,s^{-1}\,K^{-7/2}}.$$

Perpendicular to $\mathbf{B}$, each electron gyrates through $\Omega_c\tau_c \gg 1$ orbits between collisions, so the cross-field transport is suppressed by $(\Omega_c\tau_c)^{-2} \sim 10^{-18}$ in a coronal loop. To machine precision the effective conductivity is **rank-1**:

$$\boxed{\mathbf{F}_c = -\kappa_\parallel(T)\,\hat{\mathbf{b}}\,(\hat{\mathbf{b}}\cdot\nabla T), \qquad \hat{\mathbf{b}} = \mathbf{B}/|\mathbf{B}|.} \quad \text{(C6-Fc)}$$

22.2 Parallel / perpendicular decomposition

Sympy-verified the following on a general smooth $(T, \mathbf{B})$:

1. **Idempotent projector.** $\mathsf{P}_\parallel \equiv \hat{\mathbf{b}}\hat{\mathbf{b}}^{\mathrm{T}}$ satisfies $\mathsf{P}_\parallel^2 = \mathsf{P}_\parallel$ and $\mathrm{tr}\,\mathsf{P}_\parallel = 1$.
2. **Orthogonal decomposition.** $\nabla T = \mathsf{P}_\parallel\nabla T + (\mathsf{I} - \mathsf{P}_\parallel)\nabla T$ with the second piece strictly perpendicular to $\hat{\mathbf{b}}$.
3. **Anisotropy bound.** For any ∇T the flux magnitude is capped: $|\mathbf{F}_c| = \kappa_\parallel|\hat{\mathbf{b}}\cdot\nabla T| \leq \kappa_\parallel|\nabla T|$.

22.3 Kirchhoff potential (1D closed form)

In the smooth isothermal-field (or B-aligned) limit the flux admits a **gradient-form** potential:

$$\boxed{\mathbf{F}_c = -\nabla\left[\tfrac{2}{7}\,\kappa_0\,T^{7/2}\right] \quad \text{(1D / B-aligned smooth flow).}} \quad \text{(C6-Kirchhoff)}$$

This is critical for implementation: the nonlinear flux can be written as $-\kappa_0\partial_x(T^{7/2})\cdot\frac{2}{7}$, which is a **linear** central-difference operator on the Kirchhoff-transformed variable $\Theta \equiv \frac{2}{7}\kappa_0 T^{7/2}$. Sympy verifies $\partial_x(\frac{2}{7}\kappa_0 T^{7/2}) = \kappa_0 T^{5/2}\partial_x T$ identically.

22.4 Low-density collisionless quench

At $\rho \lesssim 10^{-20}\ \mathrm{g\,cm^{-3}}$ the mean-free-path exceeds the scale height and the Spitzer formula over-predicts the flux by orders of magnitude (Gruzinov-Quataert 2004; Shoda+2018a). Suzuki 2203.15280 Eq. 12 / Shoda+2020 patch this with a phenomenological cutoff:

$$\boxed{\mathbf{q}_{\mathrm{cnd}} = -\min\Big(1,\ \rho/\rho_{\mathrm{cnd}}\Big)\,(B_r/|\mathbf{B}|)\,\kappa_0\,T^{5/2}\,\partial_r T, \quad \rho_{\mathrm{cnd}} = 10^{-20}\ \mathrm{g\,cm^{-3}}.} \quad \text{(C6-quench)}$$

Do not drop the quench — unquenched Spitzer at coronal ρ drives Δt_{cond} to $\sim 10^{-4}\times$ the hyperbolic CFL (shown in Shimizu+22 Fig. 2a). The quench is a numerical device and must be on; flipping it off produces the correct physics (sharper thermal fronts) but the wall-clock cost is unmanageable.

22.5 Energy-equation coupling

The flux enters the total-energy equation conservatively:

$$\partial_t E + \nabla\cdot\big[(E+p^\star)\mathbf{v} - \mathbf{B}(\mathbf{v}\cdot\mathbf{B}) + \mathbf{F}_c\big] = 0. \quad \text{(C6-energy)}$$

Sign. Heat flows *down* the temperature gradient: $\mathbf{F}_c{\cdot}\nabla T = -\kappa_\parallel(\hat{\mathbf{b}}{\cdot}\nabla T)^2 \le 0$ (sympy-verified). Entropy production is strictly non-negative:

$$\sigma_{\text{cond}} \equiv -\frac{\mathbf{F}_c\cdot\nabla T}{T^2} = \frac{\kappa_\parallel}{T^2}(\hat{\mathbf{b}}\cdot\nabla T)^2 \ge 0. \quad \text{(C6-entropy)}$$

This is the 2nd-law certificate. Any kernel regression that flips a sign on $\mathbf{F}_c$ will show as $\sigma_{\text{cond}} < 0$ in the verification test.

22.6 Parabolic CFL (linearised FTCS)

Linearising around a background T_0 gives an effective thermal diffusivity

$$\chi_{\text{eff}} \equiv \frac{\kappa_0\, T_0^{5/2}}{\rho\, c_v}.$$

Forward-Euler + central-space applied to $\partial_t T = \chi\partial_x^2 T$ has the amplification factor

$$g(\xi) = 1 - 4\sigma\,\sin^2(\xi/2), \qquad \sigma \equiv \chi\,\Delta t\,/\,\Delta x^2. \quad \text{(C6-FTCS)}$$

Sympy-verified: worst case $\xi = \pi$ gives $g = 1 - 4\sigma$; marginal stability at $\sigma = 1/2$ gives $g = -1$. For the Spitzer diffusivity the bound becomes

$$\boxed{\Delta t_{\text{cond}} \le \tfrac{1}{2}\,\min_{\text{cells}} \frac{\rho\, c_v\, \Delta x^2}{\kappa_0\, T^{5/2}}.} \quad \text{(C6-CFL)}$$

At chromospheric parameters ($T \sim 10^5\,$K, $\rho \sim 10^{-10}\,\text{g}\,\text{cm}^{-3}$, $\Delta x \sim 50\,$km) this gives $\Delta t_{\text{cond}} \sim 10^{-2}\,$s, roughly a factor of 10^3 tighter than the hyperbolic CFL. At $T \sim 2\times10^6\,$K the factor is $\sim 10^6$.

22.7 RKL2 super-time-stepping

Because the conductive CFL is so tight, all modern stellar-wind codes use **RKL2 super-time-stepping** (Meyer+2012, based on Alexiades-Amiez-Gremaud 1996). N Chebyshev sub-stages lift the explicit diffusion step up to

$$\Delta t_{\text{RKL2}} \approx \tfrac{N^2+N}{4}\,\Delta t_{\text{cond}}. \quad \text{(C6-RKL2)}$$

Per hydrodynamic step we pick $N = \lceil\sqrt{4\Delta t_{\text{hyp}}/\Delta t_{\text{cond}}}\rceil$ so that one RKL2 block spans the hydro step. The per-block cost is N conduction-flux evaluations; for $N = 20$ the conductive solver consumes $\sim 10\%$ of the wall-clock — negligible.

RKL2 is a **separate operator** applied after the hyperbolic step (operator splitting), **not** a modification of the VL2 integrator.

22.8 Implementation recipe (for `athena_mhd` future addition)

1. **Compute** T from primitive p/ρ and μ (partial-ionisation μ from §C4 Saha closure).
2. **Evaluate flux at faces**: $\mathbf{F}_c^{i\pm1/2}$ using centred differences of T and the face-averaged $\hat{\mathbf{b}}$. Key subtlety: $\hat{\mathbf{b}}$ is a unit vector — compute it by first averaging $\mathbf{B}$ to face centres, then normalising. Averaging $\hat{\mathbf{b}}$ after normalising at cell centres loses rotational symmetry.
3. **Apply quench** per (C6-quench).
4. **Add** $\nabla\cdot\mathbf{F}_c$ to the RHS of the energy equation.
5. **RKL2 sub-cycling** if $\Delta t_{\mathrm{cond}} < 0.1\,\Delta t_{\mathrm{hyp}}$; otherwise integrate in-line with the hyperbolic step.

22.9 □ Verification checkpoints

- `tests/test_athena_mhd_conduction_isothermal.cu` — sinusoidal $T(x) = T_0(1 + A\cos kx)$ on a uniform $\mathbf{B}_0 = B_0\hat{x}$ background, no flow. Lock $L^2(T - T_0)$ decay rate matches analytical $\exp(-\chi_{\mathrm{eff}}k^2t)$ to $< 1\%$ at $N = 128$.
- `tests/test_athena_mhd_conduction_anisotropy.cu` — same IC but with $\mathbf{B}_0 = B_0\hat{y}$ (perpendicular to ∇T). Assert $L^2(T - T_0)$ stays constant over 100 collision times — cross-field quench to machine precision.
- `tests/test_athena_mhd_conduction_kirchhoff.cu` — in 1D with $T(x) \propto (1 + A\cos kx)$, compare evolution of $\frac{2}{7}\kappa_0 T^{7/2}$ versus direct nonlinear $\kappa_0 T^{5/2}\partial_x T$; lock relative difference $< 10^{-12}$.
- `tests/test_athena_mhd_conduction_entropy.cu` — random $(T, \mathbf{B}, \nabla T)$ states, 100 samples; assert $\sigma_{\mathrm{cond}} \geq 0$ in every sample. Catches accidental sign flips.

Any failure on the anisotropy (perpendicular-quench) test is an immediate red flag: without rank-1 conductivity, 1D modelling of coronal loops (Aschwanden 2005 §4) overshoots T_{peak} by a factor of 2–3.

23 C7. Optically-thin radiative cooling

sympy script: `scripts/c7_optically_thin_cooling.py` **verified:** $\Lambda(T) > 0$ on piecewise power-law segments; τ_{cool} positive definite; Townsend 2009 closed-form $T(t) = [T_0^{1-\alpha} - C(1-\alpha)t]^{1/(1-\alpha)}$ numerically verified to satisfy $\mathrm{d}T/\mathrm{d}t = -CT^{\alpha}$ across 21 (α, T_0, C, t) samples (max err 0.0); $\alpha = 1$ degenerate exponential decay; logarithmic slope identity $\mathrm{d}\ln\Lambda/\mathrm{d}\ln T = \alpha$. **code checkpoints:** `athena_mhd_kernels.cu::d_cooling_source`, `athena_mhd_solver.cu::apply_radiative_cooling`, `tests/test_athena_mhd_cooling_townsend.cu`.

23.1 Regime of validity

Optically-thin cooling applies when the mean-free-path of emitted photons exceeds the local scale height, $\ell_{\mathrm{photon}} \gtrsim H_\rho$. For a coronal plasma at $T \gtrsim 10^{5.2}\,\mathrm{K}$, $\rho \lesssim 10^{-13}\,\mathrm{g\,cm^{-3}}$ this is satisfied and the spectrum is dominated by bremsstrahlung + line emission (Sutherland-Dopita 1993, henceforth SD93).

Below $T \sim 1.5 \times 10^4\,\mathrm{K}$ the chromosphere is **optically thick** to the dominant Lyman-α and Ca II lines; see §C8 for the blended thick-thin closure used in Suzuki+25 / Shimizu+22.

23.2 SD93 cooling rate

$$\boxed{Q_R(T,\rho,Z) = n_e\, n_i\, \Lambda(T,Z) \;\geq\; 0, \qquad n_e \approx n_i \approx \rho/(\mu_e m_u).} \quad \text{(C7-QR)}$$

$\Lambda(T,Z)$ is tabulated from the SD93 ionisation-equilibrium synthesis for $Z \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 3\}\, Z_\odot$ over $\log T \in [4, 8.5]$. Piecewise power-law fit:

$$\Lambda(T) \approx \Lambda_k\, (T/T_k)^{\alpha_k}, \qquad T \in [T_k, T_{k+1}]. \quad \text{(C7-piecewise)}$$

Typical slopes (SD93 Table 6, solar Z):

Regime	$\log T$	α	Physics
Chromospheric tail	4.0–4.3	$+3$ to $+5$	HI, Ca II lines
Line-dominated	4.3–7.0	-0.5 to -1	Fe, O, Si
Bremsstrahlung	7.0–8.5	$+1/2$	free-free

23.3 Cooling timescale

$$\tau_{\text{cool}} = \frac{\varepsilon_{\text{th}}}{Q_R} = \frac{p}{(\gamma-1)\, n_e n_i\, \Lambda(T)} = \frac{\mu_e^2 m_u^2 k_B T}{(\gamma-1)\, \mu m_u\, \rho\, \Lambda(T)}. \quad \text{(C7-tau)}$$

Sympy-verified $\tau_{\text{cool}} > 0$ for positive inputs.

Order-of-magnitude: at solar corona base ($T \sim 10^6\,\text{K}$, $\rho \sim 10^{-15}\,\text{g}\,\text{cm}^{-3}$, $\Lambda \sim 10^{-22.7}\,\text{erg}\,\text{cm}^3\,\text{s}^{-1}$) $\tau_{\text{cool}} \sim 10^5\,\text{s}$, comparable to a sound- crossing time over $\sim 1\,R_\odot$. At the chromospheric transition region τ_{cool} drops to $\sim 10^0\,\text{s}$ — dramatically sub-CFL — motivating operator splitting with implicit / exactly-integrable sub-cycles (Townsend 2009).

23.4 Townsend 2009 closed-form integration

If $\Lambda(T) = \Lambda_0 (T/T_0)^\alpha$ on a power-law segment and n_e, ρ vary slowly (isochoric + slow-ρ splitting), the cooling ODE

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -C\, T^\alpha, \quad C \equiv (\gamma-1)(\mu m_u/k_B)\, n_e\, \Lambda_0/T_0^\alpha > 0$$

admits the closed-form solution (Townsend 2009 Eq. 26):

$$\boxed{T(t) = \left[\, T_0^{1-\alpha} - C(1-\alpha)\, t\,\right]^{1/(1-\alpha)} \qquad (\alpha \neq 1).} \quad \text{(C7-Townsend)}$$

Numerically verified (21 samples across $\alpha \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, \frac{3}{2}, 2, 3\}$ and representative (T_0, C, t)) to satisfy $\mathrm{d}T/\mathrm{d}t + CT^\alpha = 0$ to machine precision.

Degenerate $\alpha = 1$ limit:

$$T(t) = T_0\, \exp(-Ct). \quad \text{(C7-exp)}$$

Kernel significance: on any time-step Δt_{hyp} that spans many τ_{cool}, (C7-Townsend) gives the *exact* sub-segment update — no sub-cycling required. This is the trick that makes Athena++ / PLUTO cooling kernels run at hyperbolic CFL instead of $0.1\,\tau_{\text{cool}}$.

The full Townsend scheme handles crossing segment boundaries via a **temporal evolution function** $Y(T)$ that monotonically maps $T\mapsto$ integrated time; look-up + inversion stays closed-form.

23.5 Energy-equation coupling

$$\partial_t E + \nabla\cdot\big[\text{(hydro fluxes)}\big] = -Q_R(T,\rho,Z). \quad \text{(C7-energy)}$$

Cooling is a **pure sink**: $Q_R \geq 0$ subtracts from internal energy. Sympy confirms $\mathrm{d}\ln\Lambda/\mathrm{d}\ln T = \alpha$ on each segment, so the Field 1965 isobaric stability criterion reads

$$\partial_T \Lambda\big|_p < 0 \iff \alpha < 2, \quad \text{(C7-field)}$$

satisfied almost everywhere in $\log T \in [4.3, 7.0]$. This is the root of **thermal instability** in the ISM: the warm neutral / cold neutral bistability, and the Parker-like corona / chromosphere thermal segregation.

23.6 Explicit-source CFL (fallback, no Townsend)

When Townsend integration is disabled or the $\Lambda(T)$ table is not piecewise-power-law, sub-cycle:

$$\Delta t_{\text{rad}} \leq \beta_{\text{rad}}\,\tau_{\text{cool}}, \qquad \beta_{\text{rad}} \approx 0.1. \quad \text{(C7-CFL)}$$

$\beta_{\text{rad}} = 0.1$ is the Townsend 2009 calibrated safety margin; > 0.2 gives $\gtrsim 1\%$ errors in thermal-front propagation (SD93 shock tube benchmarks).

23.7 Metallicity scaling

SD93 $\Lambda(T,Z)$ tables decompose as

$$\Lambda(T,Z) = \Lambda_{\text{H,He}}(T) + (Z/Z_\odot)\,\Lambda_{\text{metals}}(T).$$

Two-parameter bilinear table lookup $\Lambda(\log T, \log Z)$ implemented as a 64×16 device texture; cost negligible once built (Suzuki+25 approach). For $Z < 10^{-4}\,Z_\odot$ (Pop III) the metal term is $\lesssim 10^{-3}$ of the total — atomic-H-only cooling tables (Anninos+1997) suffice instead.

23.8 Implementation recipe

1. **Tabulate** $\Lambda(\log T, \log Z)$ at build time or load from disk; store on GPU as 2D texture.
2. **Each step**: compute T from primitive (requires μ from §C4).
3. **Cooling sub-stage**: either (a) Townsend closed-form on the segment, or (b) explicit sub-cycle at $\beta_{\text{rad}}\tau_{\text{cool}}$.
4. **Apply** $\Delta E = -Q_R \cdot \Delta t$ after the hyperbolic step. No change to $\rho, \mathbf{v}, \mathbf{B}$ — the cooling is pure internal-energy relaxation.

5. **Safety floor**: clip $T \geq T_{\text{floor}}$ (e.g., $T_{\text{floor}} = 0.7T_{\text{eff}}$ in Suzuki+25) to prevent numerical runaway on under-resolved thermal fronts. The floor is a **known bias**, not a bug — document it in the driver log.

23.9 □ Verification checkpoints

- `tests/test_athena_mhd_cooling_townsend.cu` — uniform box, $\Lambda(T) = \Lambda_0(T/T_0)^{-1/2}$ (bremsstrahlung), fixed ρ. Assert $T(t)$ matches (C7-Townsend) at $t = \{0.1, 1, 10\}\,\tau_{\text{cool},0}$ to $< 10^{-10}$ relative error.
- `tests/test_athena_mhd_cooling_segment_cross.cu` — initial T_0 straddles two power-law segment boundaries; lock that the integrated $T(t)$ matches piecewise Townsend to $< 10^{-8}$.
- `tests/test_athena_mhd_cooling_energy_floor.cu` — pathological $T_0 < T_{\text{floor}}$, assert T sticks at floor and no NaN.
- `tests/test_athena_mhd_cooling_field_instability.cu` — seed an isobaric T perturbation in a Field-unstable regime ($\alpha = -1$); lock linear growth rate to analytic $\sigma = |\alpha - 2|/(2\tau_{\text{cool},0})$ within 5%.

The Field-instability test is the second-law certificate: any regression that drops the Q_R sign will mistake cooling for heating and wreck this test first.

24 C8. Chromospheric (optically-thick) cooling and blended closure

sympy script: `scripts/c8_chromospheric_cooling.py` **verified:** blend weight $\xi_{\text{rad}} \in [0, 1)$ with anchor at $p = p_{\text{chr}}$; Newton cooling has closed-form exponential solution $T(t) = T_{\text{ref}} + (T_0 - T_{\text{ref}}) \exp(-t/\tau_{\text{thck}})$; GN05 scaling $\mathrm{d}\ln\tau_{\text{thck}}/\mathrm{d}\ln\rho = -1/2$; Anderson-Athay cooling is monotone in Z, saturated above ρ_{cr}; convex-combination partials $\partial_{Q_{\text{thck}}} Q_R = \xi$, $\partial_{Q_{\text{thin}}} Q_R = 1 - \xi$. **code checkpoints:** `athena_mhd_kernels.cu::d_chromo_newton_cooling`, `athena_mhd_kernels.cu::d_anderson_athay_cooling`, `athena_mhd_solver.cu::apply_blended_cooling` `tests/test_athena_mhd_chromo_relax.cu`.

24.1 Why the chromosphere needs its own closure

At $T \lesssim 1.5 \times 10^4$ K the dominant radiative loss channels (Lyman-α, Ca II H+K, Mg II h+k, Balmer continuum) are **saturated** — the emergent intensity equals the Planck function at the optical-surface temperature, so cooling becomes independent of the *local* $n_e n_i$ and instead relaxes toward the photospheric T_{eff}. The Sutherland-Dopita 1993 optically-thin rate (§C7) overestimates cooling by $10^4\times$ in this regime (Vernazza-Avrett-Loeser 1981 VAL3 atmosphere). Suzuki-group codes patch this with a two-component closure.

24.2 Shimizu+22 blending formula

$$\boxed{Q_R(T, \rho, p) = \xi_{\text{rad}}(p)\, Q_R^{\text{thck}}(T, \rho) + (1 - \xi_{\text{rad}}(p))\, Q_R^{\text{thin}}(T, \rho, Z).} \quad \text{(C8-blend)}$$

with pressure-switch blending

$$\xi_{\text{rad}}(p) = \max\!\big(0,\ 1 - p_{\text{chr}}/p\big), \qquad p_{\text{chr}} \approx 0.1\, p_{\odot}. \quad \text{(C8-xi)}$$

Physical interpretation. In the dense chromosphere $p \gg p_{\rm chr}$ gives $\xi \to 1$ (pure Newton cooling toward the photospheric temperature). In the tenuous corona $p \lesssim p_{\rm chr}$ gives $\xi \to 0$ (pure optically-thin SD93). The transition region at $T \sim 15000\,\mathrm{K}$ is where $p \sim p_{\rm chr}$ and the closure smoothly blends both channels.

Sympy-verified $\xi_{\rm rad} \in [0, 1)$ and anchors at zero at the cutoff $p = p_{\rm chr}$.

24.3 Gudiksen-Nordlund 2005 Newton cooling

$$\boxed{Q_R^{\rm thck} = \frac{e_{\rm int} - e_{\rm int}^{\rm ref}(r)}{\tau_{\rm thck}(\rho)}, \qquad \tau_{\rm thck}(\rho) = 0.1\,(\rho/\bar{\rho})^{-1/2}\,\mathrm{s},} \quad \text{(C8-GN05)}$$

with $\bar{\rho} = 1.87 \times 10^{-7}\,\mathrm{g\,cm^{-3}}$ (Shimizu+22 calibration) and $T^{\rm ref}(r) = T_\odot$ (or $T_{\rm eff}$ in generic stars).

Exponential relaxation. For fixed ρ, the ODE $\mathrm{d}T/\mathrm{d}t = -(T - T_{\rm ref})/\tau_{\rm thck}$ has the closed-form solution

$$T(t) = T_{\rm ref} + (T_0 - T_{\rm ref})\,\exp(-t/\tau_{\rm thck}). \quad \text{(C8-relax)}$$

Sympy-verified by direct substitution. The GN05 scaling $\tau_{\rm thck} \propto \rho^{-1/2}$ (sympy: log-slope $= -1/2$) lengthens relaxation in tenuous layers — crucial to keep chromospheric shock-train structures from being over-damped.

24.4 Anderson-Athay 1989 empirical chromospheric rate

An independent branch used in Suzuki-Ohnaka-Yasuda 2025 (eq. 17):

$$\boxed{Q_R^{\rm AA}(\rho, Z) = 4.5 \times 10^9\,(0.2 + 0.8\,Z/Z_\odot)\,\min(1,\,\rho/\rho_{\rm cr})\,\mathrm{erg\,cm^{-3}\,s^{-1}}, \qquad \rho_{\rm cr} = 10^{-16}\,\mathrm{g\,cm^{-3}}.} \quad \text{(C8-AA)}$$

Properties (sympy-verified): - Strictly non-negative for $\rho, Z \geq 0$. - Monotone increasing in $Z/Z_\odot$: $\partial_Z Q_{\rm AA} = 3.6 \times 10^9 \cdot \min(1, \rho/\rho_{\rm cr}) \geq 0$. - Saturates at $\rho \geq \rho_{\rm cr}$: above the critical density, the chromosphere becomes fully optically thick and the cooling rate is density-independent (a photospheric-photon property).

The Anderson-Athay form is **not** exchangeable with the GN05 Newton form — the former is a *rate* (per volume), the latter is a *relaxation toward a reference state*. The Suzuki+25 code selects AA for RGB-wind atmospheres, Shimizu+22 selects GN05 for solar wind.

24.5 Two-piece thin cooling piece

$$Q_R^{\rm thin}(T, \rho, Z) = \begin{cases} Q_R^{\rm SD93}(T, Z)\,n_e n_i, & T > 1.2 \times 10^4\,\mathrm{K} \\ Q_R^{\rm AA}(\rho, Z), & T \leq 1.2 \times 10^4\,\mathrm{K} \\ 0, & T \leq T_{\rm cut} = 0.7\,T_{\rm eff} \end{cases}. \quad \text{(C8-twopiece)}$$

$T_{\rm cut}$ is a **numerical-stability floor**, not a physical cutoff. Without it the chromospheric rate diverges at the photospheric boundary and ejects the density to $\rho \to 0$. The $0.7\,T_{\rm eff}$ threshold is Suzuki+25 calibration (Sec 2.5); at $T_{\rm cut} = 3010\,\mathrm{K}$ for α Boo.

24.6 Convex combination preserves positivity

For $\xi_{\rm rad} \in [0,1]$ and $Q_R^{\rm thck}, Q_R^{\rm thin} \geq 0$:

$$Q_R = \xi\, Q_R^{\rm thck} + (1-\xi)\, Q_R^{\rm thin} \geq 0. \quad \text{(C8-positivity)}$$

Sympy partial derivatives confirm $\partial_{Q_{\rm thck}} Q_R = \xi$ and $\partial_{Q_{\rm thin}} Q_R = 1-\xi$ — the blend is *linear in each component*, so the combined cooling inherits positivity from its parts.

24.7 Implicit update (backward-Euler Newton cooling)

Because $\tau_{\rm thck}$ can drop below the hyperbolic CFL in dense regions, the GN05 branch is usually integrated **implicitly**. For fixed ρ the single-step Backward-Euler reduces to

$$T^{n+1} = \frac{T^n + \nu\, T_{\rm ref}}{1+\nu}, \qquad \nu \equiv \Delta t/\tau_{\rm thck}, \quad \text{(C8-BE)}$$

which is **unconditionally stable** and exactly monotone toward $T_{\rm ref}$. Kernel cost: one division per cell. No GMRES / Newton-Krylov required because $Q_R^{\rm thck}$ is linear in T.

The thin-cooling branch uses Townsend 2009 (§C7 C7-Townsend); the combined step applies BE for the thick piece and Townsend for the thin piece in operator-split fashion.

24.8 Smooth blending (optional)

The piecewise $\xi_{\rm rad}$ of (C8-xi) is C^0 at $p = p_{\rm chr}$. For applications sensitive to the smoothness of Q_R (e.g., linear g-mode propagation through the transition region), a C^∞ alternative is

$$\xi_{\rm rad}^{\rm smooth}(p) = \tfrac{1}{2}\big[1 + \tanh\!\big((p - p_{\rm chr})/\Delta p\big)\big]\,. \quad \text{(C8-smooth)}$$

$\Delta p \sim p_{\rm chr}/10$ keeps the transition width small. Shimizu+22 uses the piecewise form; Suzuki+25 uses neither (pure AA below the $T = 1.2\times10^4$ threshold, no blending).

24.9 Explicit vs implicit CFL comparison

Branch	Explicit CFL	Implicit CFL
GN05 Newton (thick)	$\Delta t \leq 0.3\,\tau_{\rm thck}$	unconditional
SD93 (thin)	$\Delta t \leq 0.1\,\tau_{\rm cool}$	Townsend exact
AA chromosphere	depends on ρ; trivially sub-CFL	direct multiply

Recommendation: in the kernel, use BE for GN05 and Townsend for SD93 in an operator-split cooling pass applied after the hyperbolic step. Neither branch touches $\rho, \mathbf{v}, \mathbf{B}$ — only internal energy.

24.10 □ Verification checkpoints

- `tests/test_athena_mhd_chromo_relax.cu` — 1D isobaric atmosphere seeded at $T_0 = 6000\,\mathrm{K}$ above $T_{\mathrm{ref}} = 5770\,\mathrm{K}$ with $\tau_{\mathrm{thck}} = 0.1\,\mathrm{s}$; lock that $T(t)$ matches (C8-relax) at $t = \{0.1, 1, 10\}\,\tau_{\mathrm{thck}}$ to $< 10^{-10}$ relative error.
- `tests/test_athena_mhd_chromo_blend.cu` — 1D atmosphere spanning $T \in [3000, 2\times 10^6]\,\mathrm{K}$; assert $\xi_{\mathrm{rad}}(p)$ profile matches (C8-xi) cell-by-cell; assert smoothness / monotonicity.
- `tests/test_athena_mhd_chromo_positivity.cu` — 100 random (T, ρ, p, Z) samples, assert $Q_R \geq 0$ and no NaN.
- `tests/test_athena_mhd_chromo_floor.cu` — atmosphere cooled past $T_{\mathrm{cut}} = 0.7\,T_{\mathrm{eff}}$; assert T stays at floor and $Q_R^{\mathrm{thin}} = 0$ there.

The positivity test is the most important — it catches any sign-flip regression in the AA or SD93 branches, which would let the code "heat via radiation" and crash.

25 D1. Ideal MHD in cylindrical (R, ϕ, z) coordinates

sympy script: `scripts/d1_cylindrical_mhd.py` **verified:** displays cylindrical divergence, curl, and axisymmetric reductions. No new nontrivial identities beyond Part A; this section is the reference for D2 and D3. **code checkpoints:** `athena_mhd_cylindrical_solver.cu` (future).

25.1 Cylindrical operators

$$\nabla \cdot \mathbf{v} = \frac{1}{R}\partial_R(R v_R) + \frac{1}{R}\partial_\phi v_\phi + \partial_z v_z.$$

$$(\nabla\times\mathbf{B})_R = \frac{1}{R}\partial_\phi B_z - \partial_z B_\phi,\ (\nabla\times\mathbf{B})_\phi = \partial_z B_R - \partial_R B_z,\ (\nabla\times\mathbf{B})_z = \frac{1}{R}\left[\partial_R(R B_\phi) - \partial_\phi B_R\right].$$

25.2 Radial momentum (axisymmetric)

$$\partial_t(\rho v_R) + \frac{1}{R}\partial_R(R\rho v_R^2) + \partial_z(\rho v_R v_z) = \boxed{\frac{\rho v_\phi^2}{R}} - \partial_R p - \rho\partial_R\Phi_{\mathrm{grav}}. \quad \text{(D1-mom-R)}$$

The $\rho v_\phi^2/R$ term is the **Christoffel-symbol / centrifugal source**. It is what keeps a rotating disk in equilibrium against gravity. In a kernel, it is the most common source of subtle bugs — easy to drop when porting Cartesian kernels.

25.3 Axisymmetric radial-flux conservation

$\partial_\phi = \partial_z = 0$ with $\mathbf{B} = B_R\hat{R}$ forces $\nabla\cdot\mathbf{B} = (1/R)\partial_R(R B_R) = 0$, so $R \cdot B_R(R) = \mathrm{const}$ — the cylindrical analog of the spherical $r^2 B_r$ conservation.

25.4 □ Verification

`tests/test_mhd_cyl_centrifugal.cu` — rotating equilibrium disk with $v_\phi = \sqrt{GM/R}$. Lock $\max|v_R|/c_s < 10^{-4}$ over 10 rotations. Failure means the centrifugal source term is missing or mis-signed.

26 D2. Shearing-sheet approximation and shearing-periodic BC

sympy script: `scripts/d2_shearing_sheet_bc.py` **verified:** shearing-sheet background is steady state (Coriolis + tidal cancel on $v_\phi^{\rm bg}$); shear wrap-around time $t_{\rm shear} = L_y/(q\Omega_0 L_x)$; effective potential is anti-confining. **code checkpoints:** `athena_mhd_shearingbox_bc.cu`, `tests/test_mhd_shearingbox_static.cu`.

26.1 Hill expansion

Local Cartesian patch centred at (R_0, Ω_0), co-rotating with the disk. Linearised rotational velocity:

$$\boxed{v_\phi^{\rm bg}(x) = -q\,\Omega_0\,x, \qquad q \equiv -\frac{d\ln\Omega}{d\ln R}}. \quad \text{(D2-shear)}$$

$q = 3/2$ for Keplerian disks, $q = 2$ for rigid rotation.

26.2 Frame-rotation sources

Coriolis:

$$\mathbf{F}_{\rm Cor} = -2\,\Omega_0\hat{z}\times\mathbf{u} = 2\Omega_0(u_y\hat{x} - u_x\hat{y}).$$

Tidal (Hill):

$$\mathbf{F}_{\rm tidal} = 2q\Omega_0^2\,x\,\hat{x} = -\nabla\Phi_{\rm eff}, \qquad \Phi_{\rm eff} = -q\Omega_0^2 x^2.$$

Sympy verified that on the background $\mathbf{u}_{\rm bg} = (0, -q\Omega_0 x, 0)$, $F_{{\rm Cor},x} + F_{{\rm tidal},x} = 0$. The shearing-sheet background is an exact steady state of the force-balance.

26.3 Shearing-periodic BC

$$\boxed{\text{field}(x = L_x, y, z, t) = \text{field}(x = 0, y - q\Omega_0 L_x t, z, t).} \quad \text{(D2-BC)}$$

The y-shift $\Delta y(t) = q\Omega_0 L_x t$ grows linearly in time. After $t_{\rm shear} = L_y/(q\Omega_0 L_x)$ the shift wraps to L_y and the BC becomes pure periodic again.

Under periodic-y convention, the BC is always well-defined: take $\Delta y \bmod L_y$. **Sympy verified** $t_{\rm shear}$.

26.4 Effective potential is anti-confining

$$\frac{d^2\Phi_{\rm eff}}{dx^2} = -2q\Omega_0^2 < 0,$$

so the tidal potential alone is unstable — **the Coriolis force is what stabilises the shearing sheet**. Sympy-verified; documented to flag that dropping the Coriolis kernel would immediately disperse the disk.

26.5 □ Verification

`tests/test_mhd_shearingbox_static.cu` — initial condition $\mathbf{u} = (0, -q\Omega_0 x, 0)$, purely hydro, no perturbation. Lock max perturbation energy $< 10^{-6} \times E_0$ over $10/\Omega_0$. Any drift means Coriolis or tidal coupling is mis-implemented.

27 D3. MRI stress decomposition $\alpha_{\text{SS}} = \alpha_R + \alpha_M$

sympy script: `scripts/d3_mri_stress_decomposition.py` **verified:** Maxwell stress antisymmetric under $B_R \to -B_R$, invariant under $(B_R, B_\phi) \to (-B_R, -B_\phi)$; Reynolds stress antisymmetric under $v_R \to -v_R$; MRI dispersion vanishes at Balbus-Hawley extremum ($k_*^2 v_A^2 = 15/16\,\Omega^2$, $\gamma_{\max} = 3/4\,\Omega$). **code checkpoints:** `tests/test_mhd_mri_growth_rate.cu`, `tests/test_mhd_mri_stress_decomp.cu`.

27.1 Shakura-Sunyaev stress

$$T_{R\phi} = \underbrace{\rho v_R\, \delta v_\phi}_{\text{Reynolds}} + \underbrace{(-B_R B_\phi)}_{\text{Maxwell}}. \quad \text{(D3-stress)}$$

$$\alpha_{\text{SS}} = \frac{\langle T_{R\phi} \rangle}{\langle p \rangle} = \alpha_R + \alpha_M.$$

27.2 Suzuki 2023 sign-quadrant decomposition

Maxwell stress is: - **antisymmetric** under single flip $B_R \to -B_R$: $M_{R\phi} \to -M_{R\phi}$. - **invariant** under simultaneous flip $(B_R, B_\phi) \to (-B_R, -B_\phi)$.

This pair of symmetries is what makes the Suzuki 2305.12112 "triangle diagnostic" meaningful: the four sign quadrants $(\pm B_R, \pm B_\phi)$ carry physically distinct stress contributions. The cylindrical vs Cartesian sign-flip result in that paper (headline $[\phi \Rightarrow_R \phi]$ arrow $+6.42$ vs -1.05) is a direct consequence of this symmetry structure.

27.3 Balbus-Hawley MRI dispersion

For vertical $\mathbf{B}_0 = B_0 \hat{z}$ and Keplerian $q = 3/2$ ($\kappa^2 = \Omega^2$):

$$\omega^4 - \omega^2(\kappa^2 + 2k^2 v_A^2) + k^2 v_A^2 (k^2 v_A^2 + \kappa^2 - 4\Omega^2) = 0. \quad \text{(D3-BH-disp)}$$

Instability requires $k^2 v_A^2 < 3\Omega^2$. The most-unstable mode:

$$\boxed{k_*^2 v_A^2 = \tfrac{15}{16}\Omega^2, \qquad \gamma_{\max} = \tfrac{3}{4}\Omega.} \quad \text{(D3-MRI-max)}$$

Sympy verified by direct substitution into the dispersion polynomial. This gives the MRI timescale $\sim 4/(3\Omega) \approx 1/3$ of an orbital period.

27.4 Parseval for Maxwell

$$\frac{1}{V}\int B_R(\mathbf{x})B_\phi(\mathbf{x})\,d^3x = \sum_{\mathbf{k}} \mathrm{Re}[\hat{B}_R(\mathbf{k})\hat{B}^*_\phi(\mathbf{k})]$$

Allows FFT-based computation of α_M at arbitrary resolution.

27.5 □ Verification

`tests/test_mhd_mri_growth_rate.cu` — seed a linear B_y perturbation at k_* with amplitude $10^{-6}B_0$; lock measured growth rate matches $(3/4)\Omega$ to $< 1\%$ over $5/\Omega$.

`tests/test_mhd_mri_stress_decomp.cu` — in nonlinear saturated state, lock $\alpha_M/\alpha_R \sim 3-5$ (Stone+96, Suzuki+23 Cartesian baseline). Suzuki+23 reports $\alpha_M \approx 0.072$, $\alpha_R \approx 0.016$ for Cartesian Keplerian baseline.

28 E1. Stochastic broadband photospheric driver

> **sympy script:** `scripts/e1_stochastic_driver.py` **verified:** $\int_{\omega_{\min}}^{\omega_{\max}}(A^2/\omega)\mathrm{d}\omega = A^2\ln(\omega_{\max}/\omega_{\min})$; target-variance normalisation $A^2 = \langle\delta v^2\rangle/\ln(\omega_{\max}/\omega_{\min})$; single-sinusoid time-variance $\langle\sin^2\rangle_t = 1/2$; cross-frequency terms vanish in time average (Parseval); linearised Elsässer $\partial_t z^\pm \pm v_A\partial_r z^\pm = 0$ (pure one-way advection around uniform background); zero-gradient BC on z^- forces incoming amplitude to zero (absorbing / no reflection in WKB limit). **code checkpoints:** `athena_mhd_kernels.cu::d_photospheric_driver`, `athena_mhd_kernels.cu::d_absorbing_bc_zminus`, `tests/test_athena_mhd_driver_spectrum.cu`.

28.1 Why a broadband stochastic driver

Suzuki-group solar / red-giant wind simulations require continuous wave injection at the inner boundary to replace the granulation-driven photospheric convection. The driver must

1. **Carry the correct Poynting flux** so that the corona is heated to observed $T \sim 10^6$ K.
2. **Span a broadband spectrum** because phenomenological sub-grid turbulence (§C5) needs a realistic range of interacting frequencies to cascade.
3. **Be stochastic in phase** so that coherent resonances with the global domain modes do not build up.
4. **Pair with an absorbing BC for the incoming Alfvén wave** so that coronal material reflecting back down does not artificially amplify the driver.

28.2 Driver form (Shimizu+22 eq. 38 / 42; Suzuki+25 Sec 2.8)

The transverse Elsässer variable outgoing from the photosphere is driven as

$$\boxed{z^+_{\perp,\odot}(t) = A_\perp \sum_{N=0}^{N_{\max}} \frac{\sin(2\pi f_N t + \varphi_N)}{\sqrt{f_N}}, \qquad \varphi_N \sim U[0,2\pi).} \quad \text{(E1-driver-transverse)}$$

The longitudinal radial velocity is driven independently:

$$\delta v_{\parallel,\odot}(t) = A_\parallel \sum_{N=0}^{N_{\max}} \frac{\sin(2\pi f_N^\parallel t + \varphi_N^\parallel)}{\sqrt{f_N^\parallel}}. \quad \text{(E1-driver-long)}$$

Log-spaced sampling mimics the continuous ω^{-1} spectrum:

$$f_N = f_{\min}\,(f_{\max}/f_{\min})^{N/N_{\max}}, \qquad f_{\max} = 100\,f_{\min}. \quad \text{(E1-logspace)}$$

Suzuki-group calibration: - Transverse (Alfvén): $f_{\min} \approx 10^{-3}$ Hz, $f_{\max} \approx 10^{-2}$ Hz (16.7 min – 100 s); - Longitudinal (p-mode): $f^\parallel_{\min} \approx 3.33 \times 10^{-3}$ Hz, $f^\parallel_{\max} \approx 10^{-2}$ Hz (5 min – 100 s).

28.3 Power spectrum and normalisation

The $1/\sqrt{f_N}$ amplitude weighting reproduces a **flat-power-per- log-octave** spectrum: $P(\omega) \propto 1/\omega$. Sympy-verified:

$$\int_{\omega_{\min}}^{\omega_{\max}} \frac{A^2}{\omega}\,\mathrm{d}\omega = A^2 \ln(\omega_{\max}/\omega_{\min}). \quad \text{(E1-spectrum)}$$

Normalising to the observed target rms fluctuation (Suzuki+25 solar calibration: $\langle \delta v_\perp \rangle_\odot = 1.25\,\mathrm{km/s}$; RGB α Boo: $2.50\,\mathrm{km/s}$ via $\delta v \propto (T_{\mathrm{eff}}^4/\rho)^{1/3}$):

$$\boxed{A^2 = \frac{\langle \delta v^2 \rangle}{\ln(\omega_{\max}/\omega_{\min})}, \qquad \int P(\omega)\,\mathrm{d}\omega = \langle \delta v^2 \rangle.} \quad \text{(E1-norm)}$$

28.4 Parseval variance identity

For a sum of sinusoids with distinct frequencies and independent phases:

$$\Big\langle \Big[\sum_N A_N \sin(\omega_N t + \varphi_N)\Big]^2 \Big\rangle_t = \tfrac{1}{2} \sum_N A_N^2. \quad \text{(E1-parseval)}$$

Sympy-verified symbolically on the time integral over a common period: single-sinusoid variance is $A^2/2$, cross-terms integrate to zero. Combining with $A_N = A/\sqrt{f_N}$ gives

$$\sum_N A_N^2 = A^2 \sum_N 1/f_N \approx A^2 \ln(f_{\max}/f_{\min})$$

(discrete approximation of the continuous integral).

The factor of $1/2$ means **the driver must be pre-multiplied by** $\sqrt{2}$ to hit $\langle \delta v^2 \rangle$ — this is the source of the explicit $\sqrt{2}$ factor in the Shimizu+22 driver (their footnote to Eq. 41).

28.5 Random phases — why essential

If φ_N are deterministic, the driver is periodic with period $T_{\text{min}} = 2\pi/\gcd(\omega_N)$ (for rational ω_N ratios) or quasi-periodic. Modes commensurate with the domain resonate — in the solar wind problem this shows up as spurious peaks in the coronal temperature spectrum.

Phase regeneration frequency. Draw new φ_N each time one simulation-correlation time elapses; practical choice $\Delta t_\varphi \sim 10/f_{\text{min}}$. Alternative: draw once per run with a fixed random seed, document the seed in run metadata (Suzuki+25 approach — makes individual runs reproducible).

28.6 Absorbing BC for incoming Alfvén

The coronal run generates downward-propagating Alfvén modes that must not pile up at the photosphere. The physically-correct boundary condition is the **free-outgoing** form in Elsässer variables:

$$\boxed{\partial_r z_\perp^-\big|_{r=R_*} = 0.} \quad \text{(E1-BC)}$$

For a plane wave $z^-(r,t) = Z_0\, e^{i(kr-\omega t)}$, this forces $ik\, Z_0 = 0$ — i.e., $Z_0 = 0$ (sympy-verified via `solve`). In WKB: **zero reflection**, all incoming wave energy is absorbed.

Kernel form. Apply zero-gradient copy on the incoming Elsässer variable at the inner ghost cells:

$$z_\perp^-(r = R_* - \text{ghost}) \leftarrow z_\perp^-(r = R_* + 1\ \text{cell}).$$

Outgoing $z_\perp^+$ is clamped to the driver value (E1-driver-transverse). Both together decouple the driver from the interior reflection.

28.7 Elsässer propagation check (background dynamics)

Around a uniform static background $(\rho_0, B_{r,0})$, the linearised transverse equations reduce to **pure one-way advection** of the Elsässer variables:

$$\partial_t z^\pm \pm v_A\, \partial_r z^\pm = 0, \qquad v_A = B_{r,0}/\sqrt{\rho_0}. \quad \text{(E1-advect)}$$

Sympy-verified directly by substituting $z^\pm = v_\perp \mp B_\perp/\sqrt{\rho_0}$ into the linearised induction + momentum equations.

Consequence: the driver at $r = R_*$ couples *only* to z^+; corrections from finite background gradients (refraction, reflection, mode conversion) appear at $\mathcal{O}(\Delta r/\lambda_A)$ and are **part of the physical solution**, not BC artifacts. This is the content of Parker (1965) and Hollweg (1981) WKB treatments.

28.8 Driver implementation recipe

```
// One-time initialisation at kernel start-up
srand(seed);
for (N = 0; N < N_max; ++N) {
    f[N] = f_min * pow(f_max/f_min, double(N)/N_max);   // log-spaced
    phi[N] = 2*M_PI * rand01();                          // uniform U[0,2π)
```

```
}
double A_perp = sqrt(2) * <dv_rms> / sqrt(log(f_max/f_min));

// Per time-step at inner ghost
double zplus = 0.0;
for (N = 0; N < N_max; ++N) {
    zplus += A_perp * sin(2*M_PI*f[N]*t + phi[N]) / sqrt(f[N]);
}
// Absorbing BC: copy interior value onto incoming Elsässer ghost
zminus_ghost = zminus_interior_1cell;
// Reconstruct v_perp, B_perp from (z+, z-)
v_perp = 0.5*(zplus + zminus_ghost);
B_perp = 0.5*sqrt(rho_0) * (zminus_ghost - zplus);
```

Key points: - **Compute $A_\perp$ once**, not per-step; the log-factor is fixed. - $\sqrt{2}$ **is important**; Parseval demands it. - **Absorbing BC copies one cell inward** (not two, not zeroth-order extrapolation). Two-cell copy is fine but a zeroth-order extrapolation will produce a small-amplitude reflection. - **Low-frequency end $f_{\rm min}$ sets the run length floor**: one $f_{\rm min}^{-1}$ period must fit within the simulation, else the driver is aliased.

28.9 Special case: acoustic-only (no transverse driver)

Shimizu+22 Appendix A runs "B0V06" with $A_\perp = 0$ to isolate the contribution of longitudinal acoustic waves to coronal heating via mode-conversion. Set $A_\perp = 0$ and drive only $\delta v_\parallel$ at the 5-min p-mode band. This probes the chromospheric Alfvén generation rate via the $\partial_r \ln p$ coupling term (§B2 Elsässer source $S_{\rm refl}$).

28.10 Per-star calibration scaling

For an arbitrary star, the solar calibration is scaled via Shimizu+22 Eq. 40:

$$\langle \delta v_0 \rangle \propto (T_{\rm eff}^4/\rho_0)^{1/3}, \qquad \omega_{\rm max}^{-1} \propto c_{s,0}\, R_*^2/M_* \text{ (photospheric transit)},$$

with solar anchors $\langle \delta v_0 \rangle_\odot = 1.25\,\text{km/s}$, $(\omega_{\rm max}^{-1})_\odot = 0.3\,\text{min}$. Numbers in Suzuki+25 Table 3:

- α Boo: $\langle \delta v_0 \rangle = 2.50\,\text{km/s}$, $\omega_{\rm max}^{-1} = 150\,\text{min}$, $\omega_{\rm min}^{-1} = 1.5 \times 10^4\,\text{min}$.
- α Tau: $\langle \delta v_0 \rangle = 2.56\,\text{km/s}$, $\omega_{\rm max}^{-1} = 340\,\text{min}$, $\omega_{\rm min}^{-1} = 3.4 \times 10^4\,\text{min}$.

28.11 □ Verification checkpoints

- `tests/test_athena_mhd_driver_spectrum.cu` — build the driver time-series over $10^6 f_{\rm min}^{-1}$, compute the FFT-measured spectrum, assert log P vs log f has slope -1 ± 0.05 within the driven band.
- `tests/test_athena_mhd_driver_variance.cu` — compute $\langle [z^+]^2 \rangle$ over a multi-period sample; assert within 5% of target $\langle \delta v^2 \rangle$.
- `tests/test_athena_mhd_absorbing_bc.cu` — inject a Gaussian down-going Alfvén pulse, measure the reflected-amplitude coefficient; lock $R < 10^{-4}$ (WKB floor).
- `tests/test_athena_mhd_driver_reproducibility.cu` — same seed produces byte-identical driver output across runs; reseed gives uncorrelated phases.

The absorbing-BC test is the non-trivial one: a wrong ghost-extrapolation order shows up here as a visible reflection pulse but would pass every other spectrum / variance test silently.